STPM Physics · Revision notes

PhysiQ

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SEMESTER 1 NOTES

Mechanics & Heat · Chapters 1–12

12 chapters · 29 topics. Definitions and laws, formula tables with symbols and units, how to answer in MPM style, and the Section A traps and tricks that cost marks in the multiple-choice paper.

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Trap: a wrong option that looks right — the slip examiners rely on.

Trick: a quick check or shortcut for the multiple-choice paper.

Section A · Semester 1

SECTION A GUIDE

BEFORE YOU START: THE 6-STEP SECTION A ROUTINE

- **Read the LAST line first** — know what is being asked (a value, a ratio, a direction, "which is NOT true") before reading the data.
- **Circle the trigger words**: NOT, only, always, never, increases, decreases, ratio, "per unit", peak / r.m.s., radius / diameter.
- **Convert units immediately**: $mm^2 \rightarrow \times 10^{-6}\ m^2$, $cm \rightarrow \times 10^{-2}\ m$, $\mu = 10^{-6}$, $n = 10^{-9}$, $p = 10^{-12}$, °C → K, minutes → seconds.
- **Estimate first**: do a rough order-of-magnitude calculation (powers of ten) — usually two options are wildly off and can be removed before you do any precise arithmetic.
- **Match the answer to the distractors**: examiners build wrong options from a forgotten square, a missing factor of 2, an inverted ratio, a wrong sign, or an un-converted unit. If your answer matches a "classic slip" value, re-check that step.
- **Time-box**: about 1½ minutes per question. Mark a hard one, move on, return later. Never leave a question blank — eliminate what you can and choose.

UNIVERSAL TRICKS THAT WORK IN EVERY SEMESTER

Trick: Ratio method — for "if X is doubled/tripled, what happens to Y?", write $Y \propto$ (formula), then substitute the factors (e.g. $\frac{2}{3^2} = \frac{2}{9}$). Never calculate with real numbers when a ratio will do.

Trick: Dimension / unit check — if the unit of your final expression is not the unit asked for, it is wrong. Quick way to remove 1–2 options.

Trick: Extreme-case check — imagine $r \rightarrow \infty$, $t \rightarrow 0$ or $t \rightarrow \infty$. Does the option behave sensibly (e.g. a discharging capacitor must end at zero)?

Trick: Direction/sign questions — draw a 2-second sketch with axes. Do not trust your head for left-hand/right-hand rules or signs.

Trick: "Which statement is correct" questions — test each option separately and write ✓ / × next to it; one clearly false word (e.g. "always", "only") normally makes an option false.

Trick: Graph questions — decide first what the gradient and the area under the graph mean physically, then read the options.

SEMESTER 1 — HOTSPOTS (MECHANICS, GASES, THERMODYNAMICS)

Trap: Uncertainties always ADD (absolute for +/−, fractional for ×/÷), and a power multiplies the fractional uncertainty ($Z = A^2 \rightarrow 2 \times \frac{\Delta A}{A}$).

Trap: v–t graph: gradient = acceleration, area = displacement. "Distance" counts area below the axis as positive; "displacement" subtracts it.

Trap: Projectile top point: vertical velocity is zero, horizontal velocity is NOT, and acceleration is still g.

Trap: Third-law pairs act on different bodies — they can never "cancel" each other on one object.

Trap: Momentum is conserved in EVERY collision (isolated system); kinetic energy only in elastic ones.

Trap: Circular motion: speed constant ≠ velocity constant; there is always a centripetal acceleration; there is no "centrifugal force" on the object in an inertial frame.

Trap: Gravitation: higher orbit → SLOWER speed, LONGER period. $V = -\frac{GM}{r}$ is negative and rises towards zero as r increases.

Trap: Gases: temperature in kelvin ALWAYS; ΔU depends only on ΔT; Q is heat INTO the gas, W is work done BY the gas.

Trick: Equal-density planets: $M \propto R^3$ so $g \propto R$. Escape speed $\propto R$ as well.

Trick: Orbits: $v \propto r^{-½}$, $T \propto r^{3/2}$, $\omega \propto r^{-3/2}$. Ratio questions take 10 seconds if you remember these.

Trick: Equations of motion (uniform acceleration): choose the equation that does NOT contain the quantity you neither know nor want.

Trick: p–V graph: the adiabatic is steeper than the isothermal through the same point; the area under the curve is the work done by the gas.

Trick: Moment of inertia: distance enters as r^2, so moving mass outward has a bigger effect than adding mass.

Chapter 1

PHYSICAL QUANTITIES

1.1 Physical Quantities & Units

DEFINITIONS & LAWS

- **Base quantities** are physical quantities that are independent and cannot be defined in terms of other quantities. The seven SI base quantities and units are: length (metre, m), mass (kilogram, kg), time (second, s), electric current (ampere, A), thermodynamic temperature (kelvin, K), amount of substance (mole, mol) and luminous intensity (candela, cd).
- **Derived quantities** are quantities defined in terms of base quantities by multiplication or division; their units are derived units (e.g. N = kg m s^{-2}, J = kg m^2 s^{-2}, W = kg m^2 s^{-3}, Pa = kg m^{-1} s^{-2}).
- **Dimension** of a physical quantity is the way it is expressed in terms of the base quantities, written in square brackets, e.g. [L], [M], [T].
- **Homogeneity principle**: every term in a physically correct equation must have the same dimensions (and units).
- Prefixes: tera T 10^{12}, giga G 10^{9}, mega M 10^{6}, kilo k 10^{3}, deci d 10^{-1}, centi c 10^{-2}, milli m 10^{-3}, micro μ 10^{-6}, nano n 10^{-9}, pico p 10^{-12}.
- Dimensionless quantities (pure numbers, angles in radian, ratios, refractive index, strain) have no unit.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Speed or velocity (dimensions)	$[v] = [L][T]^{-1}$	[L] = length; [T] = time; SI unit m s^{-1}
Acceleration (dimensions)	$[a] = [L][T]^{-2}$	[L] = length; [T] = time; SI unit m s^{-2}
Force (dimensions)	$[F] = [M][L][T]^{-2}$	[M] = mass; [L] = length; [T] = time; SI unit N
Energy and work (dimensions)	$[M][L]^2[T]^{-2}$	SI unit J
Power (dimensions)	$[M][L]^2[T]^{-3}$	SI unit W
Pressure (dimensions)	$[M][L]^{-1}[T]^{-2}$	SI unit Pa (N m^{-2})
Density (dimensions)	$[M][L]^{-3}$	SI unit kg m^{-3}

Quantity / law	Formula	Symbols, units & conditions
Frequency (dimensions)	$[T]^{-1}$	SI unit Hz
Momentum (dimensions)	$[M][L][T]^{-1}$	SI unit kg m s^{-1}

UNIT CONVERSIONS

Quantity	Conversion	Working
Speed	1 km h^{-1} = $\frac{1}{3.6}$ m s^{-1} ≈ 0.278 m s^{-1}	1 km h^{-1} = $\frac{1000\text{ m}}{3600\text{ s}}$; 3600 ÷ 1000 = 3.6. So km h^{-1} ÷ 3.6 gives m s^{-1}, and m s^{-1} × 3.6 gives km h^{-1} (e.g. 72 km h^{-1} = 20 m s^{-1})
Density	1 g cm^{-3} = 10^3 kg m^{-3}	1 g = 10^{-3} kg and 1 cm^3 = 10^{-6} m^3, so $\frac{10^{-3}}{10^{-6}}$ = 10^3
Area	1 cm^2 = 10^{-4} m^2	1 cm = 10^{-2} m, so 1 cm^2 = $(10^{-2}\text{ m})^2$ = 10^{-4} m^2
Volume	1 cm^3 = 10^{-6} m^3	1 cm = 10^{-2} m, so 1 cm^3 = $(10^{-2}\text{ m})^3$ = 10^{-6} m^3

HOW TO ANSWER (MPM STYLE)

- To check an equation: write the dimensions (or SI base units) of every term, then compare both sides; state "the equation is homogeneous / dimensionally consistent" or "not homogeneous, so it is incorrect".
- To find the unit of a constant: make the constant the subject, substitute SI units of each quantity, simplify to base units, then name the unit if it has a special name.
- Give final units in SI with correct prefix conversion before substituting into any formula.
- A conclusion must be worded: homogeneity is necessary but not sufficient for correctness.

SECTION A TRAPS

Trap: Dimensional analysis can prove an equation wrong but cannot prove it correct: dimensionless factors such as ½, 2 or π are not tested, and a wrong equation can still be homogeneous.

Trap: Unconverted prefixes: 5 cm^2 is 5×10^{-4} m^2, not 5×10^{-2} m^2; 3 g cm^{-3} is 3×10^3 kg m^{-3}.

Trap: Kilogram, not gram, is the SI base unit of mass; newton, joule and watt are derived units, not base units.

TRICKS

Trick: Replace every quantity by its base-unit string and cancel; if the powers differ on the two sides, eliminate that option at once.

Trick: The argument of an exponential, sine or logarithm function must be dimensionless, so the argument has no unit, which fixes the unit of any constant inside it.

1.2 Scalars and Vectors

DEFINITIONS & LAWS

- **Scalar quantity** is a quantity that has magnitude only (e.g. distance, speed, mass, time, energy, temperature).
- **Vector quantity** is a quantity that has both magnitude and direction (e.g. displacement, velocity, acceleration, force, momentum).
- **Resultant** is the single vector that has the same effect as the vectors it replaces.
- Vectors are added by the triangle law (head to tail) or the parallelogram law (tail to tail, resultant is the diagonal from the common tail).
- **Vector subtraction**: $A - B = A + (-B)$, where $-B$ has the same magnitude as B but opposite direction.
- **Resolution** of a vector is replacing it by two perpendicular components that together have the same effect.
- **Unit vector** has magnitude 1 and gives direction only; a vector is written $A = A_x i + A_y j$.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Vector components	$A_x = A\cos\theta$; $A_y = A\sin\theta$	A = magnitude of the vector; θ = angle between A and the x-axis; A_x, A_y = components along x and y
Magnitude and direction	$A = \sqrt{A_x^2 + A_y^2}$; $\tan\theta = \frac{A_y}{A_x}$	A = magnitude; A_x, A_y = components; θ = direction angle from the x-axis
Resultant of two vectors	$R = \sqrt{A^2 + B^2 + 2AB\cos\theta}$	R = magnitude of resultant; A, B = magnitudes of the vectors; θ = angle between them (tail to tail)
Resultant of perpendicular vectors	$R = \sqrt{A^2 + B^2}$	A, B = magnitudes of the two perpendicular vectors; R = magnitude of resultant
Resultant of equal vectors	$R = 2A\cos\frac{\theta}{2}$	A = magnitude of each vector; θ = angle between them; R = magnitude of resultant
Sine rule	$\frac{a}{\sin\alpha} = \frac{b}{\sin\beta} = \frac{c}{\sin\gamma}$	a, b, c = sides of the vector triangle; α, β, γ = angles opposite a, b, c

Quantity / law	Formula	Symbols, units & conditions
Cosine rule	$c^2 = a^2 + b^2 - 2ab\cos\gamma$	a, b, c = sides of the vector triangle; γ = angle opposite c
Resultant range	$\lvert A - B\rvert \le R \le A + B$	A, B = magnitudes of the two vectors; R = magnitude of resultant

HOW TO ANSWER (MPM STYLE)

- Draw a labelled vector diagram (triangle or parallelogram) with arrows and angles before calculating.
- Resolve each vector along two chosen perpendicular axes, sum the components (ΣF_x, ΣF_y), then state $R = \sqrt{\Sigma F_x^2 + \Sigma F_y^2}$ and the angle $\theta = \tan^{-1}\frac{\Sigma F_y}{\Sigma F_x}$ with respect to a stated axis.
- State the direction of the resultant, as an angle and its reference axis, as well as its magnitude with unit.
- For a change in a vector (e.g. $\Delta v = v - u$), draw u reversed and add.

SECTION A TRAPS

Trap: Using θ from the axis in $R^2 = A^2 + B^2 + 2AB\cos\theta$: θ must be the angle between the two vectors tail to tail; head-to-tail triangle uses the cosine rule with a minus sign and the exterior angle.

Trap: Sine and cosine swapped: the component adjacent to the stated angle uses cos θ; if the angle is measured from the vertical or from an incline surface, the cos and sin roles interchange.

Trap: Distance and speed are scalars; displacement and velocity are vectors. A body in uniform circular motion has constant speed but changing velocity.

Trap: Magnitude of the vector difference of two equal-magnitude perpendicular vectors is $\sqrt{2}$ times the magnitude, not zero and not 2 times.

TRICKS

Trick: Test limiting angles: θ = 0° gives R = A + B, 90° gives $\sqrt{A^2 + B^2}$, 180° gives $\lvert A - B\rvert$; eliminate options outside this range.

Trick: Three-four-five and 5-12-13 triples save time in perpendicular resultants.

1.3 Uncertainties in Measurements

DEFINITIONS & LAWS

- **Accuracy** is the closeness of a measured value to the true (accepted) value; it is limited by systematic error.
- **Precision** is the closeness of repeated measurements to one another (the spread of readings); it is limited by random error and indicated by a small uncertainty.
- **Systematic error** is an error that makes all readings differ from the true value by the same amount in the same direction (e.g. zero error, wrongly calibrated instrument, parallax by a consistently wrong sight); it cannot be reduced by averaging, but can be corrected by checking zero or calibrating.
- **Random error** is an error that causes readings to scatter above and below the true value in an unpredictable manner (e.g. reaction time, parallax by varying sight); it is reduced by taking many readings and calculating the mean.
- **Absolute uncertainty** Δx is the uncertainty in a measurement, with the same unit; **fractional uncertainty** is $\frac{\Delta x}{x}$; **percentage uncertainty** is $\frac{\Delta x}{x} \times 100\%$.
- Uncertainty of a single reading: analogue instrument ± half of the smallest division (some scales: ± one smallest division); digital instrument ± one smallest digit.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Mean of repeated readings	$\bar{x} = \frac{\Sigma x}{n}$	$\bar{x}$ = mean value; x = individual reading; n = number of readings
Uncertainty from repeated readings	$\Delta x = \frac{x_{max} - x_{min}}{2}$	Δx = uncertainty; x_{max}, x_{min} = largest and smallest readings
Sum or difference	$Z = A \pm B$; $\Delta Z = \Delta A + \Delta B$	ΔZ, ΔA, ΔB = absolute uncertainties, which add
Product or quotient	$Z = AB$ or $\frac{A}{B}$; $\frac{\Delta Z}{Z} = \frac{\Delta A}{A} + \frac{\Delta B}{B}$	$\frac{\Delta Z}{Z}$, $\frac{\Delta A}{A}$, $\frac{\Delta B}{B}$ = fractional uncertainties, which add
Power	$Z = A^n$; $\frac{\Delta Z}{Z} = \lvert n \rvert \frac{\Delta A}{A}$	n = power; $\frac{\Delta Z}{Z}$, $\frac{\Delta A}{A}$ = fractional uncertainties
General combination	$Z = A^m \frac{B^n}{C^p}$; $\frac{\Delta Z}{Z} = \frac{\lvert m \rvert \Delta A}{A} + \frac{\lvert n \rvert \Delta B}{B} + \frac{\lvert p \rvert \Delta C}{C}$	m, n, p = powers; $\frac{\Delta A}{A}$, $\frac{\Delta B}{B}$, $\frac{\Delta C}{C}$ = fractional uncertainties
Constant factor	$Z = kA$; $\Delta Z = k\Delta A$	k = exact number; it does not add to the fractional uncertainty

- Gradient of a graph: uncertainty = (gradient of best-fit line − gradient of worst acceptable line); similarly for intercepts.

HOW TO ANSWER (MPM STYLE)

- State the result as $(x \pm \Delta x)$ unit, with Δx to 1 significant figure (2 if it begins with 1) and x to the same decimal place as Δx.
- For a derived quantity: write the relation, state the combining rule used, substitute fractional uncertainties, then convert to the absolute uncertainty.
- For "suggest how to reduce error": random error, take repeated readings and find the mean; systematic error, correct the zero error or recalibrate; reading error, use an instrument with smaller scale division or measure a larger quantity (e.g. time 20 oscillations).
- On graphs use a straight line of best fit with points balanced, large triangle for gradient.

SECTION A TRAPS

Trap: Subtracting uncertainties: uncertainties always add, even when quantities are subtracted or divided.

Trap: Forgetting the power: for $Z = A^2B$, the contribution of A is $\frac{2\Delta A}{A}$; for a radius used as r^2, percentage uncertainty doubles.

Trap: Averaging repeated readings does not remove a zero error; a precise instrument can be inaccurate, and a set of readings with small spread can still have a large systematic error.

Trap: Adding absolute uncertainties for a product or adding fractional uncertainties for a sum: match the rule to the operation.

TRICKS

Trick: Convert everything to percentage uncertainty first when the formula contains products, quotients or powers; the answer is a simple weighted sum.

Trick: Quick check: the combined uncertainty is always larger than the largest single contribution and never larger than the sum of all contributions.

Chapter 2

KINEMATICS

2.1 Linear Motion

DEFINITIONS & LAWS

- **Distance** is the total length of the path travelled (scalar); **displacement** is the shortest straight-line distance from the initial to the final position in a stated direction (vector).
- **Speed** is the rate of change of distance; **velocity** is the rate of change of displacement (vector).
- **Instantaneous velocity** is the velocity at one instant, $v = \frac{ds}{dt}$, the gradient of the displacement–time graph at that instant.
- **Acceleration** is the rate of change of velocity, $a = \frac{dv}{dt}$ (vector).
- **Free fall** is motion under gravity alone, with constant downward acceleration $g = 9.81\ m\ s^{-2}$ near the Earth's surface (air resistance neglected), independent of mass.
- A v–t graph: gradient = acceleration; area under graph = displacement (area below the t-axis is negative). An s–t graph: gradient = velocity. An a–t graph: area = change in velocity.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Velocity–time equation	$v = u + at$	u = initial velocity; v = final velocity ($m\ s^{-1}$); a = constant acceleration ($m\ s^{-2}$); t = time (s); constant acceleration in a straight line only
Displacement equation	$s = ut + \frac{1}{2}at^2$	s = displacement (m); u = initial velocity ($m\ s^{-1}$); a = constant acceleration ($m\ s^{-2}$); t = time (s)
Velocity–displacement equation	$v^2 = u^2 + 2as$	u = initial velocity; v = final velocity ($m\ s^{-1}$); a = constant acceleration ($m\ s^{-2}$); s = displacement (m)
Displacement from mean velocity	$s = \frac{1}{2}(u + v)t$	s = displacement (m); u = initial velocity; v = final velocity ($m\ s^{-1}$); t = time (s)
Displacement from final velocity	$s = vt - \frac{1}{2}at^2$	s = displacement (m); v = final velocity ($m\ s^{-1}$); a = constant acceleration ($m\ s^{-2}$); t = time (s); valid for constant acceleration in a straight line only
Average velocity	$\frac{\Delta s}{\Delta t}$	Δs = change in displacement (m); Δt = time interval (s)

Quantity / law	Formula	Symbols, units & conditions
Average speed	total distance/total time	distance in m; time in s; speed in m s^{-1}
Vertical motion	$a = -g = -9.81$ m s^{-2}	Take one direction positive (e.g. upward); g = gravitational acceleration; at the highest point v = 0

HOW TO ANSWER (MPM STYLE)

- Sketch or state the sign convention first (positive direction), then list s, u, v, a, t with signs and pick the equation that omits the unknown.
- Write: formula → substitution with units → answer with unit, direction (sign) and 3 s.f.
- For graph work, state that the displacement is the area under the v–t graph and use the shapes (triangle ½bh, trapezium ½(a + b)h); for a curved s–t graph, draw a tangent for the instantaneous velocity.
- Sketch graphs with labelled axes and units, correct shape (straight line for constant slope, parabola for s–t when a is constant, horizontal line for a–t), intercepts and turning points.

SECTION A TRAPS

Trap: Distance and displacement confused: on a v–t graph with a part below the axis, distance uses the sum of absolute areas, displacement uses the net area.

Trap: Using the equations of motion ($v = u + at$, $s = ut + \frac{1}{2}at^2$, $v^2 = u^2 + 2as$) when acceleration is not constant (curved v–t graph or a changing a–t graph).

Trap: Sign of g: for a ball thrown upward with upward positive, $a = -g$; using $+g$ gives wrong time and height. Negative acceleration does not always mean slowing down.

Trap: At the top of the motion $v = 0$ but $a = g$, not zero; a body dropped from rest falls $s = \frac{1}{2}gt^2$, distance proportional to t^2, not t.

TRICKS

Trick: Equal time intervals from rest give distances in the ratio 1 : 3 : 5 : 7 for constant acceleration.

Trick: Time up = time down for return to the same level (no air resistance); the speed on return equals the launch speed.

Trick: Sketch the v–t graph first: a straight line means constant acceleration, so the equations of motion for uniform acceleration apply; areas then give distances quickly.

2.2 Projectiles

DEFINITIONS & LAWS

- **Projectile** is an object projected into the air that moves under the action of gravity only (air resistance neglected).
- The horizontal and vertical motions are **independent**: horizontally constant velocity ($a_x = 0$); vertically constant acceleration g downward.
- The path (trajectory) of a projectile is a **parabola**.
- At the highest point the vertical component of velocity is zero, the horizontal component is unchanged, and acceleration is g downward.
- The speed at the same height on the way up and down is the same; the time of ascent equals the time of descent when landing at the launch level.
- A projectile fired horizontally has zero initial vertical velocity; its time of fall depends only on the height h, not on the horizontal speed.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Launch velocity components	$u_x = u\cos\theta;\ u_y = u\sin\theta$	u = launch speed (m s^{-1}); θ = angle above the horizontal; u_x, u_y = horizontal and vertical components
Velocity and position at time t	$v_x = u\cos\theta;\ v_y = u\sin\theta - gt;\ x = (u\cos\theta)t;\ y = (u\sin\theta)t - \frac{1}{2}gt^2$	v_x, v_y = velocity components (m s^{-1}); x, y = horizontal and vertical displacements (m); g = gravitational acceleration (m s^{-2}); t = time (s)
Vertical velocity–displacement	$v_y^2 = u^2\sin^2\theta - 2gy$	v_y = vertical velocity (m s^{-1}); y = vertical displacement (m); u = launch speed (m s^{-1}); g = gravitational acceleration (m s^{-2})
Speed	$v = \sqrt{v_x^2 + v_y^2}$	v = speed (m s^{-1}); v_x, v_y = velocity components (m s^{-1})
Angle below horizontal	$\varphi = \tan^{-1}\frac{\lvert v_y\rvert}{v_x}$	φ = angle of velocity below the horizontal; v_x, v_y = velocity components (m s^{-1})
Time to maximum height	$t = u\,\frac{\sin\theta}{g}$	t = time (s); u = launch speed (m s^{-1}); θ = launch angle; g = gravitational acceleration (m s^{-2})
Time of flight	$T = 2u\,\frac{\sin\theta}{g}$	T = time of flight (s); u = launch speed (m s^{-1}); θ = launch angle; g = gravitational acceleration (m s^{-2}); landing at the same level
Maximum height	$H = \frac{u^2\sin^2\theta}{2g}$	H = maximum height (m); u = launch speed (m s^{-1}); θ = launch angle; g = gravitational acceleration (m s^{-2})

Quantity / law	Formula	Symbols, units & conditions
Range	$R = \frac{u^2 \sin 2\theta}{g}$	R = horizontal range (m); u = launch speed (m s^{-1}); θ = launch angle; R is maximum at θ = 45°, and θ and (90° − θ) give the same range
Trajectory	$y = x \tan \theta - \frac{gx^2}{2u^2 \cos^2 \theta}$	x, y = horizontal and vertical displacements (m); u = launch speed (m s^{-1}); θ = launch angle; g = gravitational acceleration (m s^{-2})
Horizontal launch from height h	$t = \sqrt{\frac{2h}{g}}$; $x = ut$	t = time of fall (s); h = height (m); x = range (m); u = horizontal launch speed (m s^{-1}); g = 9.81 m s^{-2}

HOW TO ANSWER (MPM STYLE)

- Resolve the initial velocity first, then treat horizontal and vertical motions separately using the same time t as the link.
- State the sign convention for the vertical direction and use a = −g (or +g if downward is positive) consistently.
- Give the final velocity as magnitude and direction (angle below/above the horizontal); give distances to 3 s.f. with unit.
- For "explain" answers use: horizontal velocity is constant because no horizontal force acts; vertical acceleration is g because only weight acts.

SECTION A TRAPS

Trap: Taking the horizontal velocity as zero at the top, or the acceleration as zero at the top; only v_y = 0 there.

Trap: Using $R = \frac{u^2 \sin 2\theta}{g}$ for a launch from a height or landing on a different level; the formula is for the same level only.

Trap: Using u instead of u cos θ in the horizontal distance, or u sin θ for the horizontal component when θ is given from the vertical.

Trap: With air resistance the range, maximum height and horizontal speed decrease, and the path is no longer symmetrical (steeper on the way down).

TRICKS

Trick: Two objects, one dropped and one fired horizontally from the same height, hit the ground at the same time.

Trick: Check by limiting angles: θ = 90° gives R = 0; θ = 0° launched from a height gives time $\sqrt{\frac{2h}{g}}$.

Trick: At equal heights speeds are equal (energy conservation), so v can be found without solving for t.

Chapter 3

DYNAMICS

3.1 Newton's Laws of Motion

DEFINITIONS & LAWS

- **Newton's first law**: a body remains at rest or moves with constant velocity in a straight line unless acted on by a resultant external force. **Inertia** is the property of a body that resists any change in its state of rest or uniform motion; it is measured by mass.
- **Newton's second law**: the rate of change of momentum of a body is directly proportional to the resultant force acting on it and takes place in the direction of the force: $F = \frac{d(mv)}{dt}$.
- **One newton** is the resultant force that gives a mass of 1 kg an acceleration of 1 m s^{-2} in the direction of the force.
- **Newton's third law**: if body A exerts a force on body B, then B exerts an equal and opposite force on A. The pair acts on different bodies, is of the same type and acts along the same line at the same time.
- **Weight** is the gravitational force on a body, W = mg, directed toward the centre of the Earth; **mass** is the amount of matter and is a scalar that does not change with place.
- **Apparent weight** is the normal reaction exerted on a body, e.g. by a lift floor; it differs from the true weight when the body accelerates vertically.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Newton's second law	$F = ma$	F = ΣF = resultant force (N); m = mass (kg), constant; a = acceleration (m s^{-2})
Constant-mass form of the law	$F = \frac{mv - mu}{t}$	F = force (N); m = mass (kg); u, v = initial and final velocities (m s^{-1}); t = time (s)
Lift accelerating upward	$R = m(g + a)$	R = reaction force on the person (N); m = mass (kg); g = gravitational acceleration (m s^{-2}); a = acceleration of lift (m s^{-2})
Lift accelerating downward	$R = m(g - a)$	R = reaction force on the person (N); m = mass (kg); g = gravitational acceleration (m s^{-2}); a = acceleration of lift (m s^{-2})
Free fall of the lift	$R = 0$	R = reaction force (N); apparent weightlessness

Quantity / law	Formula	Symbols, units & conditions
Connected bodies, common acceleration	$a = \frac{(m_1 - m_2)g}{m_1 + m_2}$	a = common acceleration (m s^{-2}); m_1, m_2 = masses (kg); g = gravitational acceleration (m s^{-2}); light string over a smooth pulley
Tension in connecting string	$T = \frac{2m_1m_2g}{m_1 + m_2}$	T = tension (N); m_1, m_2 = masses (kg); g = gravitational acceleration (m s^{-2}); light string over a smooth pulley
Smooth incline	$a = g \sin\theta$; $N = mg \cos\theta$	a = acceleration down the slope (m s^{-2}); θ = angle of incline; N = normal reaction (N); m = mass (kg); g = gravitational acceleration (m s^{-2})

HOW TO ANSWER (MPM STYLE)

- Draw a free-body diagram showing all forces on one body, label each (W, N, T, f, applied force) and choose positive direction along the acceleration.
- Apply ΣF = ma to each body separately; for connected bodies add equations so internal tensions cancel.
- Write the third-law pair in words: force of A on B and force of B on A, same type, equal magnitude, opposite direction, different bodies.
- Express the result with unit (N) and direction.

SECTION A TRAPS

Trap: Weight and normal reaction on a block resting on a table are not a third-law pair: they act on the same body and are different types. The pair of weight is the pull of the block on the Earth.

Trap: Forgetting resultant: with a constant velocity ΣF = 0, so a moving body does not need a net force; a net force is needed only for acceleration.

Trap: Lift problems: moving upward at constant speed gives R = mg; acceleration direction, not velocity direction, determines R.

Trap: Inertia gives the effect on a passenger: when a bus brakes the passenger lurches forward, when it starts the passenger moves backward relative to the bus.

TRICKS

Trick: Equal-and-opposite forces on one body mean equilibrium (first law), not third law.

Trick: For a connected system, take the total mass for ΣF = (Σm)a, then find tension from one body only.

Trick: Apparent weight greater than mg means a is upward (direction of the net force), whatever the direction of motion.

3.2 Linear Momentum

DEFINITIONS & LAWS

- **Linear momentum** of a body is the product of its mass and velocity, p = mv; a vector with the direction of the velocity, unit kg m s^{-1} or N s.
- **Principle of conservation of linear momentum**: in an isolated system (no resultant external force), the total linear momentum before an interaction equals the total linear momentum after it.
- **Impulse** is the product of the force and the time for which it acts, and equals the change in linear momentum of the body; it is a vector, unit N s.
- Newton's second law (momentum form): the resultant force equals the rate of change of linear momentum.
- A **force–time graph**: the area under the graph equals the impulse (change in momentum); the average force = impulse/time of contact.
- Momentum is conserved in all collisions and explosions of an isolated system, whether elastic or inelastic.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Linear momentum	$p = mv$	p = momentum (kg m s^{-1}); m = mass (kg); v = velocity (m s^{-1})
Conservation of momentum	$\Sigma p(\text{before}) = \Sigma p(\text{after})$; $m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$	m_1, m_2 = masses (kg); u_1, u_2 = initial velocities and v_1, v_2 = final velocities (m s^{-1}); one direction taken positive
Impulse (constant force)	$J = F\Delta t = mv - mu = \Delta p$	J = impulse (N s); F = constant force (N); Δt = time (s); u, v = initial and final velocities (m s^{-1}); Δp = change in momentum (kg m s^{-1})
Impulse (varying force)	$J = \int F\,dt$	J = impulse (N s) = area under the F–t graph; F = force (N); t = time (s)
Force as rate of change of momentum	$F = \frac{\Delta p}{\Delta t} = \frac{mv - mu}{\Delta t}$	F = force (N); Δp = change in momentum (kg m s^{-1}); Δt = time interval (s); m = mass (kg); u, v = initial and final velocities (m s^{-1})
Continuous stream hitting a surface	$F = \frac{dm}{dt}(v - u)$	F = force (N); $\frac{dm}{dt}$ = mass flow rate (kg s^{-1}); u, v = velocities before and after (m s^{-1})

Quantity / law	Formula	Symbols, units & conditions
Rebound from a wall	$\Delta p = m(v + u)$	Δp = magnitude of change in momentum (kg m s^{-1}); m = mass (kg); u = speed of approach; v = speed of return (m s^{-1})
Explosion or recoil from rest	$m_1v_1 + m_2v_2 = 0$; $v_2 = -\frac{m_1v_1}{m_2}$	m_1, m_2 = masses (kg); v_1, v_2 = velocities after the explosion (m s^{-1})
Kinetic energy and momentum	$E_k = \frac{p^2}{2m}$	E_k = kinetic energy (J); p = momentum (kg m s^{-1}); m = mass (kg)

HOW TO ANSWER (MPM STYLE)

- Define the positive direction, write the momentum equation with signs, substitute and solve; give the velocity with unit and direction.
- For impulse questions use $J = \Delta p = m(v - u)$ with the sign of u and v; then $F = \frac{J}{\Delta t}$.
- For "explain" answers: state that the total momentum is conserved because there is no external resultant force on the system.
- Always say that in a force–time graph the area is the impulse and read the value with units.

SECTION A TRAPS

Trap: Treating momentum as a scalar: an object that rebounds changes momentum by m(v + u), not m(v – u) with magnitudes only.

Trap: Believing momentum is conserved only in elastic collisions; momentum is always conserved in an isolated system, kinetic energy is not.

Trap: Longer contact time gives a smaller average force for the same change of momentum (airbag, crumple zone, bending knees on landing).

Trap: Using the total energy or speed instead of momentum for recoil: for equal momentum magnitudes, the lighter body has the larger speed and the larger kinetic energy.

TRICKS

Trick: For recoil from rest, the speeds are in inverse ratio of the masses, $\frac{v_1}{v_2} = \frac{m_2}{m_1}$.

Trick: If two bodies have equal momentum magnitudes, the one of smaller mass has the larger kinetic energy ($E_k = \frac{p^2}{2m}$).

> **Trick:** Area under F–t is impulse; if the graph is a triangle, $J = \frac{1}{2}F_{max} \cdot \Delta t$.

3.3 Elastic and Inelastic Collisions

DEFINITIONS & LAWS

- **Elastic collision** is a collision in which both total linear momentum and total kinetic energy are conserved.
- **Inelastic collision** is a collision in which total linear momentum is conserved but total kinetic energy is not conserved (some is converted to heat, sound and deformation).
- **Perfectly (completely) inelastic collision** is an inelastic collision in which the bodies stick together and move with a common velocity; the maximum loss of kinetic energy consistent with momentum conservation.
- In an elastic collision in one dimension, the relative speed of approach equals the relative speed of separation.
- In a two-dimensional collision, the total momentum is conserved separately along each of two perpendicular axes.
- In an explosion, momentum is conserved and kinetic energy increases.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Conservation of momentum	$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$	m_1, m_2 = masses (kg); u_1, u_2 = velocities before collision (m s^{-1}); v_1, v_2 = velocities after collision (m s^{-1}); holds for both elastic and inelastic collisions
Elastic collision: kinetic energy	$\frac{1}{2}m_1u_1^2 + \frac{1}{2}m_2u_2^2 = \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2$	Elastic collisions only; total kinetic energy (J) is conserved
Elastic collision: velocities	$u_1 - u_2 = v_2 - v_1$	Elastic collisions only; u = velocities before, v = velocities after (m s^{-1})
Perfectly inelastic collision	$m_1u_1 + m_2u_2 = (m_1 + m_2)v$	Perfectly inelastic collision; v = common velocity after collision (m s^{-1}); masses in kg
Elastic, m_2 initially at rest	$v_1 = \frac{(m_1 - m_2)u_1}{m_1 + m_2}$; $v_2 = \frac{2m_1u_1}{m_1 + m_2}$	Elastic collision with m_2 initially at rest; v_1, v_2 = final velocities (m s^{-1}); u_1 = initial velocity of m_1 (m s^{-1})
Loss of kinetic energy	Loss = E_k(before) − E_k(after); fractional loss = $\frac{\Delta E_k}{E_k(\text{before})}$	E_k = kinetic energy (J); ΔE_k = loss of kinetic energy (J)

Quantity / law	Formula	Symbols, units & conditions
Perfectly inelastic, m_2 at rest	$\frac{\Delta E_k}{E_k} = \frac{m_2}{m_1 + m_2}$	Perfectly inelastic collision with m_2 at rest; $\frac{\Delta E_k}{E_k}$ = fractional loss of kinetic energy
Two-dimensional collisions	Σp_x before = Σp_x after; Σp_y before = Σp_y after	p = momentum (kg m s^{-1}); conservation applied along x and along y separately

HOW TO ANSWER (MPM STYLE)

- Write momentum conservation first with a sign convention, solve for the unknown velocity, then compute kinetic energies before and after to classify the collision.
- Classification wording: "kinetic energy before = kinetic energy after, so the collision is elastic" or "kinetic energy decreases, so inelastic".
- For ballistic pendulum: momentum conservation for the collision, then energy conservation (½(M + m)v^2 = (M + m)gh) for the swing.
- State the lost kinetic energy as heat, sound and deformation when explaining.

SECTION A TRAPS

Trap: Applying kinetic energy conservation to every collision; only elastic collisions conserve kinetic energy.

Trap: Equating inelastic with sticking together: only a perfectly inelastic collision has a common final velocity.

Trap: Equal masses, head-on elastic collision with one at rest: the moving body stops and the other moves off with the original velocity; this does not hold for an inelastic collision.

Trap: In ballistic pendulum, using energy conservation across the collision, which is inelastic.

TRICKS

Trick: Check whether the relative speed of separation equals the relative speed of approach; this decides elastic quickly without calculating energies.

Trick: In an elastic head-on collision with m_1 much smaller than a heavy target, m_1 rebounds with nearly the same speed; with m_1 much larger, it continues almost unchanged.

Trick: Elastic oblique collision of equal masses (target at rest): the final velocities are perpendicular.

3.4 Friction

DEFINITIONS & LAWS

- **Friction** is the force that opposes the relative motion, or the tendency of relative motion, between two surfaces in contact; it acts parallel to the surfaces.
- **Static friction** acts when there is no relative motion; it equals the applied force up to a maximum value (limiting friction) $F_{max} = \mu_s N$.
- **Kinetic (dynamic) friction** acts when the surfaces slide; it is approximately constant and independent of speed, $F = \mu_k N$, and usually less than the limiting static friction ($\mu_k < \mu_s$).
- **Coefficient of friction** is the ratio of the friction force to the normal reaction, $\mu = \frac{F}{N}$; it has no unit and depends on the nature of the surfaces.
- Laws of friction: friction is proportional to the normal reaction; it is independent of the area of contact (for the same normal reaction) and, for kinetic friction, independent of speed.
- **Normal reaction** N is the perpendicular contact force exerted by a surface, not always equal to mg.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Static friction	$F_s \le \mu_s N$; limiting friction $F_{max} = \mu_s N$	F_s = static friction (N); μ_s = coefficient of static friction; N = normal reaction (N); F_{max} = limiting friction (N)
Kinetic friction	$F_k = \mu_k N$	F_k = kinetic friction (N); μ_k = coefficient of kinetic friction; N = normal reaction (N)
Normal reaction, pulled block	$N = mg - P \sin\theta$	Block on horizontal surface pulled by force P at angle θ above the horizontal; m = mass (kg); g = acceleration of free fall (m s^{-2}); P = pulling force (N)
Horizontal motion, pulled block	$\Sigma F_x = P\cos\theta - \mu N = ma$	Block on horizontal surface pulled by force P at angle θ above the horizontal; μ = coefficient of friction; a = acceleration (m s^{-2})
Normal reaction on incline	$N = mg\cos\theta$	Incline of angle θ; m = mass (kg); g = acceleration of free fall (m s^{-2})
Weight component down slope	$mg\sin\theta$	Incline of angle θ; component of weight down the slope (N)
Point of sliding (limiting angle)	$\tan\theta = \mu_s$	On the point of sliding; θ = angle of friction; μ_s = coefficient of static friction
Sliding at constant speed	$\tan\theta = \mu_k$	Sliding down the incline at constant speed; μ_k = coefficient of kinetic friction

Quantity / law	Formula	Symbols, units & conditions
Acceleration down rough incline	$a = g(\sin\theta - \mu_k \cos\theta)$	a = acceleration down the slope (m s^{-2}); g = acceleration of free fall (m s^{-2}); θ = angle of incline; μ_k = coefficient of kinetic friction
Acceleration up slope when pushed	$a = -g(\sin\theta + \mu_k \cos\theta)$	Body pushed up the slope; negative sign indicates deceleration (m s^{-2}); other symbols as above
Stopping distance on rough level surface	$s = \frac{u^2}{2\mu g}$	Body sliding on a rough level surface; s = stopping distance (m); u = initial speed (m s^{-1}); μ = coefficient of friction; g = acceleration of free fall (m s^{-2})
Deceleration on rough level surface	$a = \mu g$	Body sliding on a rough level surface; a = deceleration (m s^{-2})

HOW TO ANSWER (MPM STYLE)

- Draw the free-body diagram with friction opposite to the (impending) relative motion and N perpendicular to the surface.
- Resolve forces parallel and perpendicular to the surface; find N first, then friction from μN, then apply ΣF = ma.
- State whether friction is static or kinetic; if the applied force is below $\mu_s N$, state that the body stays at rest and friction equals the applied force.
- Give answers with unit, and state the direction of friction.

SECTION A TRAPS

Trap: Using F = μN with N = mg when the force has a vertical component or the surface is inclined: N changes (N = mg cos θ on an incline).

Trap: Static friction is not always $\mu_s N$; it takes the value that balances the applied force, up to the limit.

Trap: Larger area of contact does not increase friction (same normal reaction); heavier body gives larger friction because N is larger.

Trap: Mixing μ_s and μ_k: the force to start the motion uses μ_s, the force to keep it moving at constant speed uses μ_k.

TRICKS

Trick: For an incline, μ is independent of mass: a body slides at the same acceleration for any mass.

Trick: Net force on a sliding body = applied force − $\mu_k N$; if the applied force does not exceed $\mu_s N$, the body stays at rest.

Trick: If tan θ is less than μ_s, the block does not slide however large its mass.

Chapter 4

WORK, ENERGY AND POWER

4.1 Work Done by a Force

DEFINITIONS & LAWS

- **Work done** by a constant force is the product of the force and the displacement in the direction of the force, W = Fs cos θ, where θ is the angle between the force and the displacement; a scalar, unit joule (J).
- **One joule** is the work done when a force of 1 N moves its point of application through 1 m in the direction of the force.
- The work done by a variable force is the area under the force–displacement graph, W = ∫F dx.
- Work is positive when the force has a component in the direction of displacement, zero when the force is perpendicular to the displacement (e.g. normal reaction, centripetal force), and negative when the force opposes the displacement (e.g. friction).
- **Work–energy theorem**: the net work done on a body equals the change in its kinetic energy.
- A force is **conservative** if the work done is independent of the path (e.g. gravity, spring force); friction is non-conservative.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Work done by a constant force	$W = Fs\cos\theta$	Constant force; W = work done (J); F = force (N); s = displacement (m); θ = angle between force and displacement
Work done against gravity	$W = mgh$	Raising a mass m (kg) through height h (m); g = acceleration of free fall (m s^{-2}); independent of the path
Work done on a spring	$W = \frac{1}{2}kx^2$ = area under the F–x graph	Spring obeying Hooke's law F = kx; work done in stretching; k = force constant (N m^{-1}); x = extension (m)
Net work (work–energy)	$W_{net} = \Sigma F \cdot s = \frac{1}{2}mv^2 - \frac{1}{2}mu^2$	W_{net} = net work done (J); m = mass (kg); u = initial speed (m s^{-1}); v = final speed (m s^{-1})
Work done against friction	$W = \mu mgs$	Level surface; μ = coefficient of friction; m = mass (kg); g = acceleration of free fall (m s^{-2}); s = distance moved (m)

- 1 kW h = 3.6×10^6 J.

HOW TO ANSWER (MPM STYLE)

- Write: W = Fs cos θ → substitute with the correct angle between force and displacement → answer in J with 3 s.f.
- For a variable force, state that the work done equals the area under the F–s graph, and calculate with geometric shapes (triangle, trapezium).
- For a force at an angle, use only the component of force along the displacement.
- Give the sign of work and state the energy converted (e.g. to heat for friction, to PE for lifting).

SECTION A TRAPS

Trap: Using the angle between the force and the vertical (or the surface) instead of the angle between the force and the displacement; the work by weight over a vertical drop h is mgh whatever the path.

Trap: Zero work does not mean zero force: the normal reaction and the tension in circular motion do no work as they are perpendicular to the displacement.

Trap: Negative work is possible: friction and a deceleration force do negative work; the sign does not denote direction in space.

Trap: Area under an F–x graph below the x-axis is negative work.

TRICKS

Trick: Work done by gravity depends only on vertical height difference, so ignore the path length on slopes and curves.

Trick: Check the unit: N m = J = kg m^2 s^{-2}; if an answer has different base units it is wrong.

Trick: Doubling the extension of a spring multiplies the work done by 4.

4.2 Conservation of Energy

DEFINITIONS & LAWS

- **Principle of conservation of energy**: energy cannot be created or destroyed; it can only be converted from one form to another, so the total energy of an isolated system is constant.
- **Kinetic energy** is the energy a body possesses due to its motion, $E_k = \frac{1}{2}mv^2$.
- **Gravitational potential energy** is the energy a body possesses due to its position in a gravitational field; near the Earth's surface the change in potential energy is $\Delta U = mg\Delta h$.

- **Elastic potential energy** is the energy stored in a deformed spring, $U = \frac{1}{2}kx^2$.
- **Conservation of mechanical energy**: if only conservative forces do work, the sum of kinetic and potential energy is constant. If friction or air resistance acts, mechanical energy decreases and is converted to heat.
- A body released from rest from height h on a smooth surface reaches the bottom with speed $v = \sqrt{2gh}$, independent of the shape of the path.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Conservation of mechanical energy	$\frac{1}{2}mu^2 + mgh_1 = \frac{1}{2}mv^2 + mgh_2$	No resistive force; m = mass (kg); u, v = initial and final speeds (m s^{-1}); h_1, h_2 = initial and final heights (m); g = acceleration of free fall (m s^{-2})
Energy with friction	$\frac{1}{2}mu^2 + mgh_1 = \frac{1}{2}mv^2 + mgh_2 + F_f d$	Initial energy = final energy + work done against friction; F_f = frictional force (N); d = distance moved against friction (m)
Spring released from natural length	$\frac{1}{2}kx^2 = \frac{1}{2}mv^2$	Spring and mass released from the natural length; k = force constant (N m^{-1}); x = extension (m); m = mass (kg); v = speed (m s^{-1})
Spring and mass, vertical motion	$\frac{1}{2}kx^2 + mgh$	Energy interchange between the elastic and gravitational terms for vertical motion
Simple pendulum speed	$v = \sqrt{2gL(1 - \cos\theta)}$	Simple pendulum released from angle θ; v = speed at the lowest point (m s^{-1}); L = length (m); g = acceleration of free fall (m s^{-2})
Loss of energy	Loss of energy = $E_{initial} - E_{final}$	E = energy (J)
Work–energy principle	Net work = ΔE_k	ΔE_k = change in kinetic energy (J)

HOW TO ANSWER (MPM STYLE)

- Identify the initial and final points, choose a zero level for potential energy, list all energy forms at both points, then equate.
- For energy lost: calculate $E_{initial} - E_{final}$ and state it is converted to heat and sound (by friction or air resistance).
- Use energy conservation when path and time are not asked, and Newton's laws when force or time is asked.
- Quote the law in words in any "state/explain" answer: total energy is constant, energy transformed from potential to kinetic, etc.

SECTION A TRAPS

Trap: Total energy is always conserved; mechanical energy (E_k + U) is conserved only when no friction or other non-conservative force does work. Options often mix these up.

Trap: Connected bodies on a string share the same speed: use $\frac{1}{2}(m_1 + m_2)v^2$.

Trap: Taking h as path length instead of vertical height on a slope: use the vertical height $h = s \sin \theta$.

Trap: At the top of a trajectory the kinetic energy is not zero when there is a horizontal velocity.

TRICKS

Trick: Speed after a fall of h from rest needs no mass: $v^2 = 2gh$; mass cancels in the energy equation.

Trick: At the lowest point of a pendulum, all potential energy has become kinetic energy.

Trick: If speeds are asked at the same level, ignore the height terms altogether.

4.3 Power and Efficiency

DEFINITIONS & LAWS

- **Power** is the rate of doing work (or the rate of energy transfer), $P = \frac{W}{t}$; unit watt (W), 1 W = 1 J s^{-1}.
- **Instantaneous power** is the power at one instant, $P = \frac{dW}{dt} = Fv$, where v is the velocity in the direction of F.
- **Average power** is the total work done divided by the total time taken.
- **Efficiency** is the ratio of the useful output energy (or power) to the total input energy (or power), usually expressed as a percentage.
- The efficiency of a real machine is less than 100% because some energy is lost as heat (friction, resistance) and sound; energy is still conserved.
- **Kilowatt-hour** is a unit of energy: the energy transferred by a power of 1 kW in 1 hour.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Power	$P = \frac{W}{t} = \frac{\Delta E}{t}$	P = power (W); W = work done (J); ΔE = energy transferred (J); t = time (s)
Power and force	$P = Fv$	F and v in the same direction; F = driving force (N); v = velocity (m s^{-1})

Quantity / law	Formula	Symbols, units & conditions
Power to raise a load	$P = mgv$; $P = \frac{mgh}{t}$	Raising a load at constant speed; m = mass (kg); g = acceleration of free fall (m s^{-2}); v = speed (m s^{-1}); h = height (m); t = time (s)
Efficiency	η = (useful energy output/energy input) × 100% = (useful power output/power input) × 100%	η = efficiency (%); energies in J; powers in W
Power balance	Input power = output power + power wasted	Powers in W
Vehicle on a slope	F = resistive force + mg sin θ; P = Fv	Vehicle at constant speed on a slope of angle θ; F = driving force (N); m = mass (kg); g = acceleration of free fall (m s^{-2}); v = speed (m s^{-1})
Power of flowing fluid	$P = \frac{1}{2}\frac{dm}{dt}v^2 = \frac{1}{2}\rho A v^3$	Flowing fluid (e.g. water jet or wind); $\frac{dm}{dt}$ = mass flow rate (kg s^{-1}); v = speed (m s^{-1}); ρ = density (kg m^{-3}); A = cross-sectional area (m^2)

- 1 kW h = 3.6×10^6 J.

HOW TO ANSWER (MPM STYLE)

- State the relation used, substitute with units in SI (convert kW to W, minutes to s), and give the answer with unit and 3 s.f.
- For efficiency problems, identify the useful output and the input: write $\eta = \frac{P_{out}}{P_{in}}$ then rearrange.
- For a vehicle at maximum speed, state that the acceleration is zero so the driving force equals the total resistive force, then P = Fv.
- For a heated or lifted mass, find the useful energy first (mgh, mcΔθ) then divide by time.

SECTION A TRAPS

Trap: Mixing the efficiency ratio and percentage: η = 0.80 is 80%; a result above 100% is impossible.

Trap: Using P = Fv with the wrong F: at constant speed F balances resistance (plus weight component on a slope), it is not the weight alone.

Trap: For the same height and time, power is directly proportional to the mass lifted ($P = \frac{mgh}{t}$); doubling the mass doubles the power.

Trap: Energy and power confusion: kW h is energy, kW is power; a larger power does not mean more work unless the time is considered.

TRICKS

Trick: Doubling the speed against a constant resistive force doubles P ($P = Fv$); if resistance $\propto v^2$ then $P \propto v^3$.

Trick: For efficiency chains (motor then pump), the overall efficiency is the product of the individual efficiencies.

Trick: Check units: $W = J\ s^{-1} = N\ m\ s^{-1} = kg\ m^2\ s^{-3}$.

Chapter 5

CIRCULAR MOTION

5.1 Angular Velocity and Centripetal Acceleration

DEFINITIONS & LAWS

- **Angular displacement** θ is the angle swept out by the radius vector from the centre of the circle, measured in radian. 1 rad is the angle subtended at the centre of a circle by an arc equal in length to the radius.
- **Angular velocity** ω is the rate of change of angular displacement, $\omega = \frac{d\theta}{dt}$ (rad s^{-1}).
- **Period** T is the time taken for one complete revolution; **frequency** f is the number of revolutions per unit time, $f = \frac{1}{T}$ (Hz).
- **Uniform circular motion** is motion in a circle at constant speed. The velocity changes continuously in direction, so the object accelerates.
- **Centripetal acceleration** is the acceleration of an object in circular motion; it is directed towards the centre of the circle, perpendicular to the velocity, and changes the direction of the velocity but not the speed.
- Every point on a rigid rotating body has the same ω and T, but the linear speed v increases with distance r from the axis.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Angular displacement	$\theta = \frac{s}{r}$	θ = angle (rad); s = arc length (m); r = radius (m); 360° = 2π rad
Angular velocity	$\omega = \frac{d\theta}{dt} = \frac{2\pi}{T} = 2\pi f$	ω = angular velocity (rad s^{-1}); T = period (s); f = frequency (Hz)
Linear speed in circular motion	$v = r\omega$	v = linear (tangential) speed (m s^{-1}); r = radius (m); ω = angular velocity (rad s^{-1})
Centripetal acceleration	$a = \frac{v^2}{r} = r\omega^2 = v\omega$	a = centripetal acceleration (m s^{-2}), directed towards the centre; v = speed (m s^{-1}); r = radius (m); ω = angular velocity (rad s^{-1})
Angular velocity (constant α)	$\omega = \omega_0 + \alpha t$	ω_0 = initial angular velocity; ω = final angular velocity (rad s^{-1}); α = constant angular acceleration (rad s^{-2}); t = time (s)

Quantity / law	Formula	Symbols, units & conditions
Angular displacement (constant α)	$\theta = \omega_0 t + \frac{1}{2}\alpha t^2$	θ = angular displacement (rad); ω_0 = initial angular velocity (rad s^{-1}); α = constant angular acceleration (rad s^{-2}); t = time (s)
Angular velocity–displacement (constant α)	$\omega^2 = \omega_0^2 + 2\alpha\theta$	ω_0, ω = initial and final angular velocities (rad s^{-1}); α = constant angular acceleration (rad s^{-2}); θ = angular displacement (rad)
Angular displacement from mean ω (constant α)	$\theta = \frac{1}{2}(\omega_0 + \omega)t$	θ = angular displacement (rad); ω_0, ω = initial and final angular velocities (rad s^{-1}); t = time (s); rotation with constant angular acceleration

HOW TO ANSWER (MPM STYLE)

- For a definition, state the quantity, "rate of change of angular displacement", and the unit rad s^{-1}.
- Convert first: rev min^{-1} → rad s^{-1} by × $\frac{2\pi}{60}$; degrees → rad by × $\frac{\pi}{180}$; cm → m.
- Write formula → substitute with units → answer to 3 s.f. with unit; state the direction of a as "towards the centre".
- To explain why a is non-zero at constant speed: the direction of the velocity changes, so the velocity changes, hence acceleration exists.

SECTION A TRAPS

Trap: Constant speed does not mean constant velocity or zero acceleration; the acceleration is $\frac{v^2}{r}$ towards the centre.

Trap: Using rev min^{-1} or degrees directly in $v = r\omega$ or $a = r\omega^2$; ω must be in rad s^{-1}. Also using diameter in place of radius.

Trap: Two points on the same rotating disc have equal ω and T but different v and a; the point further from the axis has the larger $v = r\omega$ and $a = r\omega^2$.

TRICKS

Trick: Ratio method: at fixed ω, $v \propto r$ and $a \propto r$; at fixed v, $a \propto \frac{1}{r}$. Doubling ω at fixed r gives 4 times the acceleration.

Trick: Unit check: $\frac{v^2}{r}$ gives m s^{-2}, and $r\omega^2$ gives m s^{-2} only when ω is in rad s^{-1} (s^{-1}).

5.2 Centripetal Force

DEFINITIONS & LAWS

- **Centripetal force** is the resultant force directed towards the centre of the circle that is required to make an object move in a circular path; it is perpendicular to the velocity and does no work.
- Centripetal force is not a separate force; it is provided by tension, friction, gravitational force, normal reaction or a component of these forces.
- **Newton's first law**: a body remains at rest or moves with constant velocity unless acted on by a resultant force; hence a body in circular motion needs a resultant force towards the centre.
- Explain the "outward" sensation: it is due to inertia of the body, which tends to continue in a straight line (tangent); there is no real outward force on the body in an inertial frame.
- If the centripetal force is removed (string breaks), the object moves along the tangent at constant velocity.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Centripetal force	$F = \frac{mv^2}{r} = mr\omega^2 = mv\omega$	F = centripetal force (N); m = mass (kg); v = speed (m s^{-1}); r = radius (m); ω = angular velocity (rad s^{-1})
Horizontal circle on a string / conical pendulum	$T\sin\theta = \frac{mv^2}{r}$; $T\cos\theta = mg$; $\tan\theta = \frac{v^2}{rg}$	T = tension (N); θ = angle of string to the vertical; m = mass (kg); v = speed (m s^{-1}); r = radius (m); g = gravitational acceleration (m s^{-2})
Level bend (friction provides F)	$\mu mg \geq \frac{mv^2}{r}$; $v_{max} = \sqrt{\mu gr}$	μ = coefficient of friction; m = mass (kg); v_{max} = maximum speed without skidding (m s^{-1}); r = radius of bend (m)
Banked bend, no friction	$\tan\theta = \frac{v^2}{rg}$	θ = angle of banking; v = speed (m s^{-1}); r = radius (m); g = gravitational acceleration (m s^{-2})
Vertical circle, top	$T + mg = \frac{mv^2}{r}$	T = tension (N); m = mass (kg); v = speed at the top (m s^{-1}); r = radius (m)
Vertical circle, bottom	$T - mg = \frac{mv^2}{r}$	T = tension (N); m = mass (kg); v = speed at the bottom (m s^{-1}); r = radius (m)
Minimum speed at top of vertical circle	$v = \sqrt{gr}$	Condition T = 0 at the top; v = minimum speed (m s^{-1}); r = radius (m); g = gravitational acceleration (m s^{-2})
Top of a convex bridge	$mg - N = \frac{mv^2}{r}$	N = normal reaction (N); m = mass (kg); v = speed (m s^{-1}); r = radius of curvature (m); loop or bridge
Bottom of a dip	$N - mg = \frac{mv^2}{r}$	N = normal reaction (N); m = mass (kg); v = speed (m s^{-1}); r = radius of curvature (m); loop or bridge

HOW TO ANSWER (MPM STYLE)

- Draw a free-body diagram, identify which force(s) point towards the centre, and write "resultant force towards the centre = $\frac{mv^2}{r}$".
- Resolve forces vertically (equilibrium, no vertical acceleration) and horizontally (centripetal) for conical pendulum and banked-bend problems.
- For vertical circles state the sign convention: towards the centre positive, so the weight is added at the top and subtracted at the bottom.
- For an explanation of skidding: the required centripetal force $\frac{mv^2}{r}$ exceeds the maximum friction μmg, so the car moves tangentially outward.

SECTION A TRAPS

Trap: Centrifugal force is not an extra force on the body in an inertial frame; do not include $\frac{mv^2}{r}$ as an additional force in the free-body diagram.

Trap: Weight and tension at the top of a vertical circle both act towards the centre, so $T = \frac{mv^2}{r} - mg$; at the bottom $T = \frac{mv^2}{r} + mg$ (tension is largest at the bottom).

Trap: Fixed ω: $F \propto r$. Fixed v: $F \propto \frac{1}{r}$. Mixing these up gives the inverted-ratio option.

Trap: In $v_{max} = \sqrt{\mu gr}$ the mass cancels; a heavier car does not have a larger safe speed.

TRICKS

Trick: Doubling v quadruples F; doubling r at fixed v halves F; at fixed ω doubling r doubles F.

Trick: Centripetal force does no work because $F \perp v$; so the kinetic energy of uniform circular motion is constant.

Chapter 6

GRAVITATION

6.1 Newton's Law of Gravitation

DEFINITIONS & LAWS

- **Newton's law of gravitation**: every point mass in the universe attracts every other point mass with a force that is directly proportional to the product of their masses and inversely proportional to the square of the distance between them.
- The gravitational force is always attractive, acts along the line joining the two masses, and the two forces form an action–reaction pair (equal magnitude, opposite direction) on the two bodies.
- A uniform spherical body acts as if its mass were concentrated at its centre, so r is measured between centres.
- **Kepler's first law**: the orbit of a planet is an ellipse with the Sun at one focus.
- **Kepler's second law**: the line joining a planet to the Sun sweeps out equal areas in equal times.
- **Kepler's third law**: the square of the period of a planet is directly proportional to the cube of its mean orbital radius (semi-major axis), $T^2 \propto r^3$.
- **Weight** is the gravitational force exerted on a body by the Earth; $g = \frac{GM}{R^2}$ at the surface.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Newton's law of gravitation	$F = \frac{GMm}{r^2}$	F = gravitational force (N); $G = 6.67\times10^{-11}$ N m^2 kg^{-2}; M, m = masses (kg); r = separation of the centres (m)
Surface gravitational field strength	$g = \frac{GM}{R^2}$; $GM = gR^2$	g = gravitational field strength at the surface (N kg^{-1}); M = mass of planet (kg); R = radius of planet (m)
Planet of density ρ	$M = \frac{4}{3}\pi R^3\rho$; $g = \frac{4}{3}\pi G\rho R$	M = mass (kg); R = radius (m); ρ = density (kg m^{-3}); G = gravitational constant
Circular orbit	$\frac{GMm}{r^2} = \frac{mv^2}{r} = mr\omega^2$; $T^2 = \frac{4\pi^2}{GM}r^3$	M = mass of central body (kg); m = orbiting mass (kg); r = orbital radius (m); v = orbital speed (m s^{-1}); ω = angular velocity (rad s^{-1}); T = period (s)
Kepler's third law	$\frac{T_1^2}{T_2^2} = \frac{r_1^3}{r_2^3}$	T_1, T_2 = periods (s); r_1, r_2 = orbital radii (m); two orbits about the same body

Quantity / law	Formula	Symbols, units & conditions
Mass of the central body	$M = \frac{4\pi^2 r^3}{GT^2}$	M = mass of central body (kg); r = orbital radius (m); T = period (s); G = gravitational constant
Density method	$M = \rho \frac{4}{3}\pi R^3$; $g \propto R$	M = mass (kg); ρ = density (kg m^{-3}); R = radius (m); $g \propto R$ holds for equal density

HOW TO ANSWER (MPM STYLE)

- State the law with "product of masses" and "square of the distance"; quote $F = \frac{GMm}{r^2}$ and define every symbol.
- In calculations convert km → m before squaring, and use r from the centre of the planet (R + h for a satellite at height h).
- For ratio questions write the ratio of formulae, cancel constants, then substitute (e.g. $\frac{F_2}{F_1} = \frac{m_2}{m_1}(\frac{r_1}{r_2})^2$).
- For Kepler's third law show $\frac{T^2}{r^3}$ is constant, then substitute; keep T and r in consistent units on both sides.

SECTION A TRAPS

Trap: Forgotten square: doubling r reduces F to one quarter (not one half); halving r gives four times F.

Trap: Using the surface-to-surface distance or the height above the surface as r instead of the distance between centres.

Trap: Equal-density planets: $M \propto R^3$, so $g \propto R$ (not R^2 or $\frac{1}{R^2}$); $g \propto \frac{1}{R^2}$ applies only for a fixed mass.

Trap: Thinking the heavier body exerts the larger force; the two forces are equal in magnitude.

TRICKS

Trick: Ratio method: $F \propto \frac{m_1 m_2}{r^2}$; write $F_2 = F_1 \times$ (mass factors) $\times$ (1/distance factor)^2.

Trick: Kepler III shortcut: $T \propto r^{3/2}$; if r is 4 times, T is 8 times; if r is 9 times, T is 27 times.

6.2 Gravitational Field and Potential

DEFINITIONS & LAWS

- **Gravitational field** is a region of space in which a mass experiences a gravitational force.
- **Gravitational field strength** g at a point is the gravitational force per unit mass acting on a small test mass placed at that point; g is a vector directed towards the mass producing the field.
- **Gravitational potential** V at a point is the work done per unit mass by an external force in bringing a small test mass from infinity to that point; V = 0 at infinity, so V is negative everywhere in the field of an attractive mass, and it is a scalar.
- **Gravitational potential energy** U of a mass m at a point is the work done by an external force in bringing the mass from infinity to that point, U = mV.
- Field strength is the negative of the potential gradient, $g = -\frac{dV}{dr}$; the area under a g–r graph gives $-\Delta V$.
- **Escape speed** is the minimum speed at which a body must be projected from the surface of a planet so that it just reaches infinity, where its speed is zero.
- An equipotential surface is a surface on which all points have the same potential; no work is done moving a mass along it, and g is perpendicular to it.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Gravitational field strength	$g = \frac{GM}{r^2}$	g = field strength (N kg^{-1}); M = mass of the source (kg); r = distance from centre (m); valid for $r \geq R$ (outside a spherical body)
Gravitational potential	$V = -\frac{GM}{r}$	V = gravitational potential (J kg^{-1}); M = mass of source (kg); r = distance from centre (m)
Gravitational potential energy	$U = -\frac{GMm}{r}$	U = potential energy (J); M = mass of source (kg); m = mass in the field (kg); r = distance from centre (m)
Field and potential gradient	$g = -\frac{dV}{dr}$	g = field strength (N kg^{-1}); V = potential (J kg^{-1}); r = distance (m)
Change in potential energy	$\Delta U = m(V_2 - V_1) = GMm(\frac{1}{r_1} - \frac{1}{r_2})$	ΔU = change in potential energy (J); V_1, V_2 = potentials (J kg^{-1}) at r_1, r_2 (m)
Change in potential energy near surface	$\Delta U = mg\Delta h$	m = mass (kg); g = gravitational field strength (N kg^{-1}); Δh = change in height (m); near the surface

Quantity / law	Formula	Symbols, units & conditions
Escape speed	$\frac{1}{2}mv^2 - \frac{GMm}{R} = 0$; $v_{esc} = \sqrt{\frac{2GM}{R}} = \sqrt{2gR}$	v_{esc} = escape speed (m s^{-1}); M = mass of planet (kg); R = radius (m); g = surface field strength (N kg^{-1}); for the Earth about 1.1×10^4 m s^{-1}
Energy conservation between two points	$\frac{1}{2}mv_1^2 - \frac{GMm}{r_1} = \frac{1}{2}mv_2^2 - \frac{GMm}{r_2}$	v_1, v_2 = speeds (m s^{-1}) at distances r_1, r_2 (m); m = mass (kg); M = mass of source (kg)
Several masses	$V = \Sigma V$	V = total potential (J kg^{-1}); scalar sum of the potentials due to each mass

- The field strength due to several masses is the vector sum of the individual fields.

HOW TO ANSWER (MPM STYLE)

- Define g as "force per unit mass" and V as "work done per unit mass in bringing a small mass from infinity"; mention V = 0 at infinity.
- For two masses, add fields as vectors (direction by components) and potentials as scalars (with sign); the neutral point is where the resultant g is zero.
- For escape speed, equate the kinetic energy to the magnitude of the potential energy: $\frac{1}{2}mv^2 = \frac{GMm}{R}$; the mass of the projectile cancels.
- Sketch V against r: labelled axes, V negative, starting at $-\frac{GM}{R}$ at r = R and rising towards zero (never positive) as $r \to \infty$; g against r: positive magnitude decreasing as $\frac{1}{r^2}$.

SECTION A TRAPS

Trap: V is always negative and increases towards zero as r increases; moving a mass away from the Earth increases its potential energy (U becomes less negative).

Trap: Using $\Delta U = mg\Delta h$ for large heights; it is valid only when g is nearly constant ($\Delta h \ll R$). Otherwise use $GMm(\frac{1}{r_1} - \frac{1}{r_2})$.

Trap: $V \propto \frac{1}{r}$ but $g \propto \frac{1}{r^2}$; doubling r halves V and quarters g. V is a scalar (J kg^{-1}); g is a vector (N kg^{-1}).

Trap: Escape speed does not depend on the mass or the direction of projection of the body, only on M and R of the planet; $v_{esc} = \sqrt{2}$ × orbital speed at the surface, not 2 ×.

TRICKS

Trick: Unit check: J kg^{-1} = m^2 s^{-2} for V; N kg^{-1} = m s^{-2} for g.

Trick: Neutral point between two masses M_1 and M_2: $\frac{GM_1}{x^2} = \frac{GM_2}{(d-x)^2}$, i.e. the point lies closer to the smaller mass and $\frac{x}{d-x} = \sqrt{\frac{M_1}{M_2}}$.

6.3 Satellite Motion

DEFINITIONS & LAWS

- **Satellite** in a circular orbit is moving under the gravitational force of the central body, which provides the centripetal force: $\frac{GMm}{r^2} = \frac{mv^2}{r}$.
- **Geostationary satellite** is a satellite that stays above a fixed point on the Earth's equator: it orbits in the equatorial plane, in the same direction as the Earth's rotation (west to east), with period 24 h (same angular velocity as the Earth), at a fixed height of about 3.6×10^7 m above the surface ($r \approx 4.2\times10^7$ m).
- **Apparent weightlessness**: in orbit the satellite and its contents have the same acceleration towards the Earth (free fall), so there is no contact (normal) force between the astronaut and the floor, although the gravitational force still acts.
- **Kepler's third law**: $T^2 \propto r^3$ for orbits about the same central mass.
- The gravitational force on the satellite is perpendicular to its velocity in a circular orbit, so it does no work and the speed and total energy remain constant.
- A satellite in a lower orbit has a greater speed and a smaller period; to move to a higher orbit the satellite gains total energy (E becomes less negative).

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Orbital speed	$v = \sqrt{\frac{GM}{r}}$	v = orbital speed (m s^{-1}); M = mass of central body (kg); r = R + h = orbital radius (m)
Period and mass of central body	$T = \frac{2\pi r}{v}$; $T^2 = \frac{4\pi^2}{GM}r^3$; $M = \frac{4\pi^2 r^3}{GT^2}$	T = period (s); r = orbital radius (m); v = orbital speed (m s^{-1}); M = mass of central body (kg)
Angular velocity and acceleration	$\omega = \sqrt{\frac{GM}{r^3}}$; $a = g_r = \frac{GM}{r^2}$	ω = angular velocity (rad s^{-1}); a = centripetal acceleration = field strength g_r at radius r (m s^{-2})

Quantity / law	Formula	Symbols, units & conditions
Kinetic energy of satellite	$K = \frac{GMm}{2r}$	K = kinetic energy (J); m = satellite mass (kg); M = mass of central body (kg); r = orbital radius (m)
Potential energy of satellite	$U = -\frac{GMm}{r}$	U = potential energy (J); m = satellite mass (kg); M = mass of central body (kg); r = orbital radius (m)
Total energy of satellite	$E = K + U = -\frac{GMm}{2r}$; $E = -K$; $U = -2K$	E = total energy (J); K = kinetic energy (J); U = potential energy (J)
Orbital speed near the surface	$v = \sqrt{gR} \approx 7.9\times10^{3}\ \text{m s}^{-1}$	v = orbital speed (m s^{-1}); g = surface field strength (N kg^{-1}); R = radius of planet (m)
Escape speed from orbital speed	$v_{esc} = \sqrt{2}\ v_{orbit}$	v_{esc} = escape speed (m s^{-1}); v_{orbit} = orbital speed (m s^{-1})
Energy to raise a satellite	$\Delta E = \frac{GMm}{2}\left(\frac{1}{r_1} - \frac{1}{r_2}\right)$	ΔE = energy needed (J); r_1, r_2 = initial and final orbital radii (m); m = satellite mass (kg); M = mass of central body (kg)
Geostationary radius	$r^3 = \frac{GMT^2}{4\pi^2}$	r = orbital radius (m); T = 24 h = 8.64×10^{4} s; M = mass of Earth (kg)

HOW TO ANSWER (MPM STYLE)

- Start with "gravitational force provides the centripetal force" and equate $\frac{GMm}{r^2}$ to $\frac{mv^2}{r}$ (or $mr\omega^2$); the satellite mass cancels.
- Always use r = R + h in the formulas; substitute M and G or use $GM = gR^2$ when g and R are given.
- State the three conditions for geostationary orbit: equatorial plane, period 24 h, same direction as the Earth's spin (west to east).
- For weightlessness, explain with free fall: satellite and astronaut have the same acceleration, so the reaction force is zero; do not say gravity is zero.

SECTION A TRAPS

Trap: "Weightless" does not mean zero gravity: g at orbital height is still large (about 8–9 N kg^{-1} at 400 km).

Trap: A higher orbit has a slower speed and longer period; the speed does not increase with r.

Trap: The total energy of a bound satellite is negative, $E = -\frac{GMm}{2r}$; do not use $+\frac{GMm}{2r}$, and do not take E as equal to K.

Trap: Orbital speed and period do not depend on the satellite mass; a heavier satellite at the same r has the same v and T.

Trap: Using the height h above the surface in place of r = R + h in $v = \sqrt{\frac{GM}{r}}$ or Kepler's law.

TRICKS

Trick: Ratio method: $v \propto r^{-1/2}$, $T \propto r^{3/2}$, $\omega \propto r^{-3/2}$; doubling r reduces v by $\frac{1}{\sqrt{2}}$.

Trick: Energy shortcut: K = |E| and U = 2E, so if r doubles, K and |E| halve.

Chapter 7

ROTATION OF RIGID BODIES

7.1 Torque and Angular Acceleration

DEFINITIONS & LAWS

- **Torque (moment of a force)** about an axis is the product of the force and the perpendicular distance from the axis to the line of action of the force.
- **Couple** is a pair of equal and opposite parallel forces whose lines of action do not coincide; it produces rotation without translation, and its torque is the product of one force and the perpendicular distance between the two forces.
- **Newton's second law of rotation**: the angular acceleration of a rigid body about a fixed axis is directly proportional to the resultant torque and inversely proportional to its moment of inertia.
- Torque is a vector (clockwise/anticlockwise sense must be stated); a force whose line of action passes through the axis has zero torque.
- **Angular acceleration** α is the rate of change of angular velocity, $\alpha = \frac{d\omega}{dt}$ (rad s^{-2}).

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Torque of a force	$\tau = Fr\sin\theta = F_{\perp}r = Fd$	τ = torque (N m); F = force (N); r = distance from axis (m); θ = angle between F and r; $F_{\perp}$ = component of F perpendicular to r; d = r sinθ = perpendicular distance (m)
Couple	$\tau = Fd$	τ = torque of the couple (N m); F = magnitude of each force (N); d = perpendicular separation of the two forces (m)
Torque and angular acceleration	$\tau = I\alpha$	τ = torque (N m); I = moment of inertia (kg m^2); α = angular acceleration (rad s^{-2})
Linear–rotational links	$s = r\theta$; $v = r\omega$; $a_t = r\alpha$	s = arc length (m); v = tangential speed (m s^{-1}); a_t = tangential acceleration (m s^{-2}); θ = angle (rad); ω = angular velocity (rad s^{-1}); α = angular acceleration (rad s^{-2}); r = radius (m)
Equations of rotational motion	$\omega = \omega_0 + \alpha t$; $\theta = \omega_0 t + \frac{1}{2}\alpha t^2$; $\omega^2 = \omega_0^2 + 2\alpha\theta$	ω_0, ω = initial and final angular velocities (rad s^{-1}); θ = angular displacement (rad); t = time (s); constant α (rad s^{-2})

Quantity / law	Formula	Symbols, units & conditions
Wheel or pulley with a hanging mass	$mg - T = ma$; $Tr = I\alpha$; $a = r\alpha$	m = hanging mass (kg); T = tension (N); a = linear acceleration of mass (m s^{-2}); r = radius of wheel (m); I = moment of inertia (kg m^2); α = angular acceleration (rad s^{-2})

HOW TO ANSWER (MPM STYLE)

- Draw the force and the axis, mark the perpendicular distance d, then write $\tau = F\,d$; state clockwise or anticlockwise.
- For $\tau = I\alpha$ problems, find the net torque about the axis (sum of torques with sign), then $\alpha = \frac{\tau}{I}$; add a linear equation for each hanging mass.
- Convert cm → m and rev min^{-1} → rad s^{-1} before substituting; give τ in N m and α in rad s^{-2}.
- For a couple, state that the resultant force is zero but the resultant torque is Fd.

SECTION A TRAPS

Trap: Using the distance along the rod without sinθ; the perpendicular distance is r sinθ (or use F sinθ). If F is perpendicular to the rod, sinθ = 1.

Trap: Torque is not measured in joules for this subtopic; unit is N m (do not mistake it for energy).

Trap: In pulley problems, using T = mg for the string tension; the hanging mass accelerates, so T = m(g − a).

Trap: Using F = ma for rotation without I: a body with larger I has smaller α for the same torque.

TRICKS

Trick: Limiting case: if the line of action passes through the pivot, d = 0 and τ = 0.

Trick: Ratio method: for the same τ, $\alpha \propto \frac{1}{I}$; doubling the radius of gyration ($I \propto r^2$) quarters α.

7.2 Moment of Inertia

DEFINITIONS & LAWS

- **Moment of inertia** I of a body about an axis is the sum of the products of the mass of each particle and the square of its perpendicular distance from the axis, $I = \Sigma mr^2$; it is the measure of the body's resistance to angular acceleration (rotational inertia).
- Moment of inertia depends on the mass, on how the mass is distributed about the axis, and on the position of the axis; the same body has different I about different axes.
- Moment of inertia is the rotational analogue of mass: $\tau = I\alpha$ corresponds to $F = ma$.
- A body with its mass distributed further from the axis has a larger I and so is harder to start or stop rotating.
- A flywheel has most of its mass at the rim to give a large I, so that it resists changes in angular speed (stores rotational kinetic energy).

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Point mass	$I = mr^2$	I = moment of inertia (kg m^2); m = mass (kg); r = perpendicular distance from the axis (m)
System of masses	$I = \Sigma m_i r_i^2 = m_1 r_1^2 + m_2 r_2^2 + \ldots$	I = moment of inertia (kg m^2); m_i = mass of the i-th particle (kg); r_i = its perpendicular distance from the axis (m)
Thin hoop or ring	$I = MR^2$	M = mass (kg); R = radius (m); about its axis
Solid disc or cylinder	$I = \frac{1}{2}MR^2$	M = mass (kg); R = radius (m); about its axis
Solid sphere	$I = \frac{2}{5}MR^2$	M = mass (kg); R = radius (m); about a diameter
Thin spherical shell	$I = \frac{2}{3}MR^2$	M = mass (kg); R = radius (m)
Thin rod about its centre	$I = \frac{1}{12}ML^2$	M = mass (kg); L = length (m); axis through the centre
Thin rod about one end	$I = \frac{1}{3}ML^2$	M = mass (kg); L = length (m); axis through one end
Radius of gyration	$I = Mk^2$	k = radius of gyration (m); M = mass (kg); I = moment of inertia (kg m^2)

HOW TO ANSWER (MPM STYLE)

- Identify the axis first; calculate the distance r of each mass from that axis (perpendicular distance), square it, multiply by the mass, and add.
- For a composite body, add the individual I values about the same axis; show every term with units kg m^2.

- For comparing bodies of the same mass and radius, give the order by mass distribution: hoop > disc > solid sphere ($\frac{I}{MR^2}$ = 1, 0.5, 0.4).
- Link to τ = Iα when asked for the angular acceleration of the body.

SECTION A TRAPS

Trap: The distance in Σmr^2 is measured from the axis, not from the centre of the body or from the other mass; a mass lying on the axis contributes zero.

Trap: Doubling the distance of a mass from the axis multiplies its contribution by 4 (r^2), not by 2.

Trap: Using the formula for the wrong axis: rod about the centre is $\frac{ML^2}{12}$, about the end is $\frac{ML^2}{3}$ (four times larger); disc about its own axis is $\frac{1}{2}MR^2$.

Trap: Moment of inertia of a rigid body is not a fixed property of the body; it is different about different axes and changes if the mass distribution is changed.

TRICKS

Trick: Keep the order I(hoop) > I(disc) > I(solid sphere) for the same M and R, since less mass at the rim lowers I.

Trick: Unit check: I has unit kg m^2; a result in kg m^{-2} or kg m indicates a missing or extra power of r.

7.3 Angular Momentum and Rotational Kinetic Energy

DEFINITIONS & LAWS

- **Angular momentum** L of a rigid body about an axis is the product of its moment of inertia and its angular velocity, L = Iω (kg m^2 s^{-1}); for a point mass L = mvr.
- **Principle of conservation of angular momentum**: the total angular momentum of a system remains constant if no resultant external torque acts on the system.
- **Rotational kinetic energy** is the energy a rotating body possesses because of its angular velocity, $\frac{1}{2}I\omega^2$.
- Newton's second law in terms of angular momentum: the resultant torque is the rate of change of angular momentum, $\tau = \frac{dL}{dt}$.
- When the moment of inertia of a system decreases (e.g. a skater draws in the arms), ω increases so that L is constant; the rotational kinetic energy increases because work is done by the skater (internal forces).

- In a collision or coupling of rotating bodies about the same axis (inelastic), angular momentum is conserved but kinetic energy is lost.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Angular momentum	$L = I\omega = mvr$	L = angular momentum (kg m^2 s^{-1}); I = moment of inertia (kg m^2); ω = angular velocity (rad s^{-1}); mvr applies to a point mass in circular motion
Torque and angular momentum	$\tau = \frac{dL}{dt} = I\frac{d\omega}{dt} = I\alpha$	τ = torque (N m); α = angular acceleration (rad s^{-2}); I constant
Conservation of angular momentum	$I_1\omega_1 = I_2\omega_2$	I = moment of inertia (kg m^2); ω = angular velocity (rad s^{-1}); no external torque
Coupling of two rotating bodies	$I_1\omega_1 + I_2\omega_2 = (I_1 + I_2)\omega$	I_1, I_2 = moments of inertia (kg m^2); ω_1, ω_2 = initial angular velocities (rad s^{-1}); ω = common final angular velocity (rad s^{-1})
Rotational kinetic energy	$K = \frac{1}{2}I\omega^2 = \frac{L^2}{2I}$	K = rotational kinetic energy (J); I = moment of inertia (kg m^2); ω = angular velocity (rad s^{-1}); L = angular momentum (kg m^2 s^{-1})
Rolling without slipping	$v = r\omega$; $K = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$	v = speed of centre (m s^{-1}); r = radius (m); ω = angular velocity (rad s^{-1}); K = total kinetic energy (J); m = mass (kg); I = moment of inertia (kg m^2)
Rolling down an incline	$mgh = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$	h = height of incline (m); starts from rest; m = mass (kg); v = speed at the bottom (m s^{-1}); ω = angular velocity (rad s^{-1})
Solid cylinder rolling down an incline	$v = \sqrt{\frac{4gh}{3}}$	v = speed at the bottom (m s^{-1}); h = height (m); g = gravitational acceleration (m s^{-2}); from rest
Work done by a torque	$W = \tau\theta$	W = work done (J); τ = torque (N m); θ = angle turned (rad)
Power of a torque	$P = \tau\omega$	P = power (W); τ = torque (N m); ω = angular velocity (rad s^{-1})

HOW TO ANSWER (MPM STYLE)

- State the principle of conservation of angular momentum with the condition "no resultant external torque" before using $I_1\omega_1 = I_2\omega_2$.
- Find I for each stage ($I = \Sigma mr^2$), substitute with ω in rad s^{-1}, and give the final ω with unit rad s^{-1} (convert to rev min^{-1} only if asked).

- When asked for the change in kinetic energy, calculate $\frac{1}{2}I\omega^2$ before and after and state the source (work done by the person/internal forces) or the loss (inelastic).
- For rolling, include both translational and rotational kinetic energy; state $v = r\omega$ as the no-slip condition.

SECTION A TRAPS

Trap: Only L is conserved when I changes; $\frac{1}{2}I\omega^2$ is not conserved. If I decreases by half, ω doubles and K doubles ($K = \frac{L^2}{2I}$).

Trap: Angular momentum is conserved only with zero resultant external torque; do not claim it for a body acted on by a net torque.

Trap: Rolling: total kinetic energy is greater than $\frac{1}{2}mv^2$; a body with a smaller $\frac{I}{MR^2}$ (solid sphere) reaches the bottom of the incline first and has the larger v, independent of m and R.

Trap: Forgetting to convert rev min^{-1} to rad s^{-1} before using $\frac{1}{2}I\omega^2$; this gives an error of a factor $(\frac{60}{2\pi})^2$.

TRICKS

Trick: Skater: $I_2 = \frac{I_1}{n}$ gives $\omega_2 = n\omega_1$ and $K_2 = nK_1$; use the ratio directly without the numbers.

Trick: Unit check: $\frac{1}{2}I\omega^2$ gives kg m^2 s^{-2} = J; if the unit is not J, ω has not been in rad s^{-1}.

Chapter 8

STATICS

8.1 Equilibrium of Forces

DEFINITIONS & LAWS

- **Translational equilibrium**: a body is in translational equilibrium when the resultant force acting on it is zero; it is at rest or moves with constant velocity.
- **Rotational equilibrium**: a body is in rotational equilibrium when the resultant torque about any axis is zero.
- **Conditions for equilibrium of a rigid body (static equilibrium)**: (1) the vector sum of all forces is zero, and (2) the sum of the moments about any point is zero.
- **Triangle of forces**: if three coplanar forces acting at a point are in equilibrium, they can be represented in magnitude and direction by the three sides of a triangle taken in order.
- For three non-parallel forces in equilibrium, the lines of action are coplanar and meet at one point (concurrent).
- A couple is two equal and opposite parallel forces; a body acted on by a couple alone is in translational equilibrium but not in rotational equilibrium.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Translational equilibrium	$\Sigma F = 0$; $\Sigma F_x = 0$ and $\Sigma F_y = 0$	ΣF = resultant force (N); F_x, F_y = components along x and y axes; for coplanar forces
Rotational equilibrium	$\Sigma \tau = 0$	τ = torque (N m); about any chosen point; Σ clockwise moments = Σ anticlockwise moments
Lami's theorem	$\frac{F_1}{\sin \alpha} = \frac{F_2}{\sin \beta} = \frac{F_3}{\sin \gamma}$	F_1, F_2, F_3 = three forces in equilibrium (N); α, β, γ = angles opposite F_1, F_2, F_3 (the angles between the other two forces)
Resolving a force	$F_x = F\cos\theta$; $F_y = F\sin\theta$	F = force (N); θ = angle of F to the horizontal; F_x = horizontal component (N); F_y = vertical component (N)
Weight	$W = mg$	W = weight (N); m = mass (kg); g = gravitational field strength (N kg^{-1}); acts vertically downwards through the centre of gravity
Friction at a rough surface	$f \leq \mu N$; $f = \mu N$ (limiting)	f = friction force (N); μ = coefficient of friction; N = normal reaction (N); equality holds at the limiting value

HOW TO ANSWER (MPM STYLE)

- Draw a clear free-body diagram showing every force (weight, tension, normal reaction, friction) with arrows and angles.
- Resolve horizontally and vertically, write $\Sigma F_x = 0$ and $\Sigma F_y = 0$, solve; state the answer with unit and direction.
- For a rigid body, take moments about the point where the most unknown forces act so that they have zero moment, and state "Σ clockwise moments = Σ anticlockwise moments".
- Use the triangle of forces/Lami's theorem for exactly three forces; the closed triangle must be drawn head-to-tail.

SECTION A TRAPS

Trap: Zero resultant force does not guarantee zero resultant torque (a couple); both conditions are needed for static equilibrium.

Trap: Using the wrong angle in resolution: when the angle is given with the vertical, the horizontal component is F sinθ, not F cosθ.

Trap: In Lami's theorem the sine is of the angle opposite that force (the angle between the other two forces), not the angle the force makes with the vertical.

Trap: Tension is not equal to the weight when the string is at an angle; T cosθ = W for two symmetric strings gives $T = \frac{W}{2\cos\theta}$.

TRICKS

Trick: Check: if two forces are equal and opposite and colinear, the third force must be zero; for three forces the largest force is opposite the largest angle.

Trick: Taking moments about the pivot removes the pivot's unknown reaction from the equation; saves one simultaneous equation.

8.2 Moments and Centre of Gravity

DEFINITIONS & LAWS

- **Moment of a force** about a point is the product of the force and the perpendicular distance from the point to the line of action of the force.

- **Principle of moments**: for a body in rotational equilibrium, the sum of the clockwise moments about any point is equal to the sum of the anticlockwise moments about the same point.
- **Centre of gravity** of a body is the point through which the whole weight of the body appears to act.
- **Centre of mass** is the point at which the whole mass of the body can be considered to be concentrated; it coincides with the centre of gravity when the gravitational field is uniform.
- The centre of gravity of a uniform regular body is at its geometric centre; for a freely suspended body it lies vertically below the point of suspension.
- **Stable equilibrium**: a body returns to its original position after a small displacement; the centre of gravity rises on displacement. **Unstable**: the body moves further away; the centre of gravity falls. **Neutral**: the body stays in the new position; the centre of gravity stays at the same height.
- A body topples when the vertical line through its centre of gravity falls outside its base.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Moment of a force	Moment = Fd	F = force (N); d = perpendicular distance to the line of action (m); moment in N m
Principle of moments	ΣF d (clockwise) = ΣF d (anticlockwise)	F = force (N); d = perpendicular distance (m); about the same point
Centre of gravity of a system of masses	$\bar{x} = \frac{\Sigma mx}{\Sigma m}$	$\bar{x}$ = position of the centre of gravity (m); m = mass (kg); x = position of mass m (m)
Uniform beam	W = mg	W = weight (N); m = mass (kg); g = gravitational field strength (N kg^{-1}); weight acts at the midpoint
Toppling on a slope	$\tan\theta > \frac{w}{h}$; $\tan\theta = \frac{\frac{w}{2}}{\frac{h}{2}}$	Uniform block of base width w (m) and height h (m) on a slope of angle θ; topples when $\tan\theta > \frac{w}{h}$; the line through the centre of gravity passes the lower edge at $\tan\theta = \frac{\frac{w}{2}}{\frac{h}{2}}$
Toppling on a horizontal surface	$P\,h \geq W\,\frac{w}{2}$	P = horizontal force (N) applied at height h (m); W = weight (N); w = base width (m); block topples when this holds
Reaction forces on a beam	$R_1 + R_2 = W$ + other loads	R_1, R_2 = reactions at the two supports (N); W = weight of beam (N); translational equilibrium

HOW TO ANSWER (MPM STYLE)

- State the pivot chosen, then "taking moments about P" and write clockwise = anticlockwise with each force × perpendicular distance.

- Mark the weight of a uniform beam at its midpoint, and the reaction at each support; show the distances on the diagram.
- Combine ΣF = 0 with moments to find unknown reactions or tensions; give each answer in N with the sense stated.
- For stability, write that the vertical line through the centre of gravity falls inside/outside the base; mention a low centre of gravity and a wide base increase stability.

SECTION A TRAPS

Trap: Using the length along the beam as the distance for an inclined force; the perpendicular distance (or F sinθ) must be used.

Trap: Forgetting the weight of the uniform beam itself at its midpoint when taking moments.

Trap: Stability: a lower centre of gravity and a wider base increase stability; the mass of the object does not by itself decide whether it topples.

Trap: Pivot at the wrong point in the equation: all moments in one equation must be about the same point; the force at the pivot has zero moment.

TRICKS

Trick: Choose the pivot where an unknown force acts to eliminate that force from the equation.

Trick: Block on a slope: the critical angle satisfies $\tan\theta = \frac{w}{h}$, so a tall narrow block (small $\frac{w}{h}$) topples at a smaller angle.

Chapter 9

DEFORMATION OF SOLIDS

9.1 Stress, Strain and Young's Modulus

DEFINITIONS & LAWS

- **Stress** is the force acting normally per unit cross-sectional area of a material.
- **Strain** is the ratio of the extension to the original length; it has no unit.
- **Young modulus** is the ratio of tensile (or compressive) stress to the corresponding strain, within the limit of proportionality.
- **Hooke's law**: the extension of a material is directly proportional to the applied force provided the limit of proportionality is not exceeded.
- **Limit of proportionality** is the point on the graph beyond which stress is no longer proportional to strain. **Elastic limit** is the maximum stress beyond which the material does not return to its original length when the force is removed.
- **Yield point** is the point at which the material extends considerably (plastic flow) with little or no increase in stress. **Ultimate tensile stress** is the maximum stress the material can withstand.
- A **ductile** material shows a large plastic region before fracture (copper, mild steel); a **brittle** material fractures just after the elastic region with almost no plastic deformation (glass, cast iron); a **polymeric** material (rubber) shows a large non-linear extension and hysteresis.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Stress	$\sigma = \frac{F}{A}$	σ = stress (Pa = N m^{-2}); F = normal force (N); A = cross-sectional area (m^2)
Strain	$\varepsilon = \frac{x}{L_0}$	ε = strain (no unit); x = extension (m); L_0 = original length (m)
Young modulus	$E = \frac{\sigma}{\varepsilon} = FL_0/(Ax)$	E = Young modulus (Pa); E is the gradient of the linear part of a σ–ε graph
Force constant	$F = kx;\ k = \frac{EA}{L_0}$	k = force constant (N m^{-1}); valid below the limit of proportionality; E = Young modulus (Pa); A = cross-sectional area (m^2); L_0 = original length (m)
Area of a circular wire	$A = \frac{\pi d^2}{4} = \pi r^2$	d = diameter (m); r = radius (m); A = cross-sectional area (m^2)
Extension of a wire under a given load	$x \propto \frac{L_0}{d^2}$	x = extension (m); L_0 = original length (m); d = diameter (m)

HOW TO ANSWER (MPM STYLE)

- Define stress and strain in words, then give the formula with symbols and units; write the strain as a pure number or percentage.
- Calculation: state $E = FL_0/(Ax)$ → convert mm to m, and use $A = \frac{\pi d^2}{4}$ with the diameter (not the radius) → substitute → answer with unit (Pa or N m^{-2}) to 3 s.f.
- Sketch of a σ–ε graph for a ductile metal: label axes (stress/Pa, strain), mark the limit of proportionality, elastic limit, yield point, ultimate tensile stress and fracture point; the straight line passes through the origin.
- Comparing materials: the steeper initial gradient means the larger Young modulus (stiffer); the larger maximum stress means stronger.

SECTION A TRAPS

Trap: Using the diameter as the radius in $A = \pi r^2$, or forgetting to convert mm^2 to m^2 ($1\ mm^2 = 10^{-6}\ m^2$); this gives an answer wrong by a factor of 4 or 10^6.

Trap: Confusing stiffness with strength: a stiffer material (larger gradient) is not necessarily the stronger one (larger ultimate tensile stress).

Trap: Reading E from a point on the curved part of the graph or from the area under the graph; E is the gradient of the initial straight line only.

TRICKS

Trick: For the same load and material, $x \propto \frac{L_0}{d^2}$. Doubling the diameter divides the extension by 4; doubling the length doubles the extension.

Trick: Unit check: Pa = N m^{-2} has the same dimensions as J m^{-3}, so energy density and stress share a unit.

9.2 Elastic Energy

DEFINITIONS & LAWS

- **Elastic potential energy (strain energy)** is the energy stored in a material when it is deformed by a force within the elastic limit.
- The work done in stretching a wire equals the area under the force–extension graph; this is stored as elastic potential energy if the elastic limit is not exceeded.
- **Energy density** is the elastic potential energy stored per unit volume of the material.

- Beyond the elastic limit, the work done is not wholly stored: part is dissipated as heat during plastic deformation, so the unloading line is not the loading curve (hysteresis for rubber: the area enclosed by the loop is the energy dissipated as heat).

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Elastic potential energy	$U = \frac{1}{2}Fx = \frac{1}{2}kx^2 = \frac{F^2}{2k}$	U = energy stored (J); F = force (N); x = extension (m); k = force constant (N m^{-1}); valid for the linear region only
Elastic energy in terms of Young modulus	$U = \frac{1}{2}\frac{EA}{L_0}x^2$	E = Young modulus (Pa); A = cross-sectional area (m^2); L_0 = original length (m); x = extension (m)
Energy density	$\frac{U}{V} = \frac{1}{2}\sigma\varepsilon = \frac{\sigma^2}{2E} = \frac{1}{2}E\varepsilon^2$	unit J m^{-3}; V = AL_0 = volume (m^3); σ = stress (Pa); ε = strain; E = Young modulus (Pa)
Work done for a non-linear curve	Work done = area under the F–x graph	count squares or use trapezium; F = force (N); x = extension (m)
Extra energy stored when load increases	$\Delta U = \frac{1}{2}(F_1 + F_2)(x_2 - x_1)$	for a linear graph; load increases from F_1 to F_2 (N); x_1, x_2 = corresponding extensions (m)

HOW TO ANSWER (MPM STYLE)

- Define elastic potential energy, then state U = ½Fx (area under the F–x graph) before substituting; convert extension to metres.
- For a stretched wire: find x from $x = \frac{FL_0}{AE}$ first, then U = ½Fx; give the answer in J to 3 s.f.
- Energy conversion: equate the loss in gravitational potential energy (or kinetic energy) to $\frac{1}{2}kx^2$ and state the assumption that no energy is lost.
- For hysteresis or a non-linear graph, state that the energy equals the area under the curve or the area enclosed by the loop, as appropriate.

SECTION A TRAPS

Trap: Using U = Fx instead of ½Fx; the factor ½ arises because the force increases from zero to F.

Trap: Using ½Fx or $\frac{1}{2}kx^2$ with the final values when the graph is curved (beyond the limit of proportionality); the area under the actual graph must be used.

Trap: Doubling the extension of the same wire multiplies U by 4 ($U \propto x^2$), not by 2; doubling the force also gives 4 times the energy.

TRICKS

Trick: For two springs or wires carrying the same force, $U = \frac{F^2}{2k}$: the smaller k stores more energy. With the same extension, $U = \frac{1}{2}kx^2$: the larger k stores more.

Trick: Unit check: $\frac{1}{2}\sigma\varepsilon$ has the unit Pa = J m^{-3}, so it is energy per unit volume, not total energy.

Chapter 10

GASES

10.1 Ideal Gas Equation

DEFINITIONS & LAWS

- **Ideal gas** is a gas that obeys the equation of state pV = nRT at all pressures and temperatures.
- **Boyle's law**: for a fixed mass of gas at constant temperature, the pressure is inversely proportional to the volume. **Charles's law**: at constant pressure, the volume is directly proportional to the thermodynamic temperature. **Pressure law**: at constant volume, the pressure is directly proportional to the thermodynamic temperature.
- **Mole** is the amount of substance that contains as many elementary entities as there are atoms in 0.012 kg of carbon-12, i.e. the Avogadro constant N_A of entities.
- A real gas behaves like an ideal gas at low pressure and high temperature (molecules far apart, intermolecular forces and molecular volume negligible).
- **Dalton's law**: the total pressure of a mixture of gases equals the sum of the partial pressures of its components.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Ideal gas equation (moles)	$pV = nRT$	p = pressure (Pa); V = volume (m^3); n = amount of substance (mol); R = 8.31 J mol^{-1} K^{-1}; T = thermodynamic temperature (K)
Ideal gas equation (molecules)	$pV = NkT$	N = number of molecules; $k = \frac{R}{N_A} = 1.38\times10^{-23}$ J K^{-1} (Boltzmann constant); T = thermodynamic temperature (K)
Amount of substance	$n = \frac{N}{N_A} = \frac{m}{M}$	$N_A = 6.02\times10^{23}$ mol^{-1}; m = mass (kg); M = molar mass (kg mol^{-1})
Gas law for fixed mass	$\frac{p_1V_1}{T_1} = \frac{p_2V_2}{T_2}$	for a fixed mass of gas; T in kelvin
Temperature conversion	$T\ (K) = \theta\ (°C) + 273$	T = thermodynamic temperature (K); θ = Celsius temperature (°C); 273.15 may be used instead of 273
Density of an ideal gas	$\rho = \frac{pM}{RT}$	from $pV = \frac{m}{M}RT$; ρ = density (kg m^{-3}); M = molar mass (kg mol^{-1})

HOW TO ANSWER (MPM STYLE)

- Write the equation first (pV = nRT), then convert: temperature to K, volume to m^3 (1 cm^3 = 10^{-6} m^3, 1 litre = 10^{-3} m^3), pressure to Pa.
- Make the quantity the subject, substitute with units, and give the answer with unit to 3 s.f.
- For a change of state of a fixed mass, use $\frac{p_1V_1}{T_1} = \frac{p_2V_2}{T_2}$ and state which quantity is constant when a law is quoted.
- Explanation of deviation: state that real gases deviate at high pressure and low temperature because the molecular volume and intermolecular forces are no longer negligible.

SECTION A TRAPS

Trap: Substituting the temperature in °C instead of K; the result can be wrong by a large factor and sometimes gives negative values.

Trap: Using the molar mass in g mol^{-1} with R = 8.31 J mol^{-1} K^{-1}, which requires kg mol^{-1}; or using the volume in cm^3 or litres without conversion to m^3.

Trap: Mixing up R and k: R is used with the number of moles n, k with the number of molecules N.

TRICKS

Trick: When only p, V and T change for a fixed mass, never compute n: use the ratio form $\frac{p_1V_1}{T_1} = \frac{p_2V_2}{T_2}$, and the units of p and V then cancel.

Trick: At constant volume, doubling the pressure doubles the Kelvin temperature, not the Celsius temperature.

10.2 Kinetic Theory and Molecular Speeds

DEFINITIONS & LAWS

- Assumptions of the kinetic theory of an ideal gas: (1) a gas consists of a very large number of identical molecules in continuous random motion; (2) the volume of the molecules is negligible compared with the volume of the container; (3) there are no intermolecular forces except during collisions; (4) collisions between molecules and with the walls are perfectly elastic; (5) the duration of a collision is negligible compared with the time between collisions; (6) Newton's laws of motion apply.
- **Pressure** of a gas is caused by the molecules colliding with the walls: the rate of change of momentum of the molecules per unit area of the wall.

- **Root-mean-square speed** c_{rms} is the square root of the mean of the squares of the speeds of the molecules.
- The average translational kinetic energy of a molecule is directly proportional to the thermodynamic temperature, and is independent of the type of gas.
- **Degrees of freedom** f is the number of independent ways in which a molecule can possess energy. Principle of equipartition of energy: each degree of freedom has an average energy of ½kT per molecule.
- A monatomic gas has f = 3 (translational), a diatomic gas has f = 5 (3 translational + 2 rotational) at ordinary temperatures, and a polyatomic (non-linear) gas has f = 6.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Pressure from kinetic theory	$p = \frac{1}{3}\rho\langle c^2\rangle = \frac{1}{3}(Nm/V)\langle c^2\rangle$	ρ = density (kg m^{-3}); N = number of molecules; m = mass of one molecule (kg); V = volume (m^3); $\langle c^2\rangle$ = mean square speed ($m^2\ s^{-2}$)
Root-mean-square speed	$c_{rms} = \sqrt{\langle c^2\rangle} = \sqrt{\frac{3kT}{m}} = \sqrt{\frac{3RT}{M}} = \sqrt{\frac{3p}{\rho}}$	M = molar mass (kg mol^{-1}); m = mass of one molecule (kg); T = temperature (K); k = Boltzmann constant (J K^{-1}); R = molar gas constant (J mol^{-1} K^{-1}); p = pressure (Pa); ρ = density (kg m^{-3})
Mean translational kinetic energy per molecule	$\langle E_k\rangle = \frac{1}{2}m\langle c^2\rangle = \frac{3}{2}kT$	for the translational motion of one molecule (J)
Mean energy per molecule	$\frac{f}{2}kT$	f = number of degrees of freedom; k = Boltzmann constant (J K^{-1}); T = temperature (K)
Internal energy of an ideal gas	$U = \frac{f}{2}nRT = \frac{f}{2}NkT$	n = number of moles; N = number of molecules; f = number of degrees of freedom; U in J
Total translational kinetic energy	$\frac{3}{2}nRT$	n moles of gas; total translational kinetic energy in J; T = temperature (K)
Pressure and kinetic energy	$pV = \frac{1}{3}N\langle E_k\rangle$	N = number of molecules; $\langle E_k\rangle$ = mean translational kinetic energy per molecule (J)

HOW TO ANSWER (MPM STYLE)

- For a derivation or statement of $p = \frac{1}{3}\rho\langle c^2\rangle$, list the assumptions first, then use the change of momentum per collision ($2mc_x$), the rate of collisions and $\langle c_x^2\rangle = \frac{\langle c^2\rangle}{3}$.
- For c_{rms}: write $c_{rms} = \sqrt{\frac{3RT}{M}}$, convert M to kg mol^{-1} and T to K, substitute and give the answer in m s^{-1} to 3 s.f.

- For a comparison of two gases at the same temperature: state that the average kinetic energy per molecule is the same, but $c_{rms} \propto \frac{1}{\sqrt{M}}$, so the lighter gas has the larger c_{rms}.
- Explain a rise in pressure when the temperature rises at constant volume: the molecules move faster, collide with the walls more frequently and with greater change of momentum per collision.

SECTION A TRAPS

Trap: Stating that all molecules have the same speed; the molecules have a range of speeds, and only the rms (and mean) speed is quoted.

Trap: Quoting $\frac{3}{2}$kT as the internal energy per molecule of a diatomic gas: the translational part is $\frac{3}{2}$ kT, but the total energy per molecule is $\frac{5}{2}$kT.

Trap: Using the molar mass in g mol^{-1} in $c_{rms} = \sqrt{\frac{3RT}{M}}$, or doubling c_{rms} when T is doubled: c_{rms} increases by a factor of $\sqrt{2}$ only.

TRICKS

Trick: $c_{rms} \propto \sqrt{\frac{T}{M}}$: doubling T raises c_{rms} by $\sqrt{2}$; at the same T the c_{rms} ratio of two gases is $\sqrt{\frac{M_2}{M_1}}$.

Trick: Same temperature means the same average translational kinetic energy per molecule, whatever the gas, so only the molecular mass decides the speed.

Chapter 11

THERMODYNAMICS

11.1 First Law of Thermodynamics

DEFINITIONS & LAWS

- **First law of thermodynamics**: the increase in the internal energy of a system is equal to the heat supplied to the system minus the work done by the system on its surroundings (conservation of energy).
- **Internal energy** of a system is the sum of the random kinetic energy and the potential energy of its molecules; for an ideal gas there is no intermolecular potential energy, so U depends only on the thermodynamic temperature.
- **Heat** is the energy transferred between a system and its surroundings because of a temperature difference. **Work** is the energy transferred when the gas expands or is compressed against an external pressure.
- **Molar heat capacity at constant volume** C_V is the heat required to raise the temperature of one mole of gas by 1 K at constant volume; **at constant pressure** C_p is the heat required for the same rise at constant pressure; C_p is greater than C_V because additional heat is needed for the work done in expansion.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
First law of thermodynamics	$\Delta U = Q - W$	ΔU = change in internal energy (J); Q = heat supplied to the gas (J), negative when heat leaves the gas; W = work done by the gas (J), negative when the gas is compressed
Change in internal energy	$\Delta U = \frac{f}{2} nR\Delta T$	for an ideal gas, whatever the process; f = number of degrees of freedom; n = number of moles; ΔT = temperature change (K)
Work done by gas	$W = p\Delta V$; W = area under the p–V graph	$W = p\Delta V$ at constant pressure; the area under the p–V graph holds in general; p = pressure (Pa); ΔV = volume change (m^3)
Heat at constant volume	$Q = nC_V\Delta T$	C_V = molar heat capacity at constant volume (J mol^{-1} K^{-1}); n = number of moles; ΔT = temperature change (K)
Heat at constant pressure	$Q = nC_p\Delta T$	C_p = molar heat capacity at constant pressure (J mol^{-1} K^{-1}); n = number of moles; ΔT = temperature change (K)

Quantity / law	Formula	Symbols, units & conditions
Molar heat capacities	$C_V = \frac{f}{2}R$; $C_p = C_V + R$	Mayer's relation: $C_p = C_V + R$; R = molar gas constant (J mol^{-1} K^{-1}); f = number of degrees of freedom
Ratio of heat capacities	$\gamma = \frac{C_p}{C_V} = \frac{f+2}{f}$	γ = ratio of molar heat capacities (no unit); f = number of degrees of freedom

HOW TO ANSWER (MPM STYLE)

- State the first law with the meaning and sign of every symbol before substituting: Q positive for heat absorbed, W positive for work done by the gas.
- Decide the signs from the wording: compressed → W negative; heat released or lost → Q negative; temperature rises → ΔU positive.
- Calculation: $\Delta U = \frac{f}{2}nR\Delta T$ first, then find the missing quantity from $\Delta U = Q - W$; give the unit J and the sign (increase or decrease, absorbed or released).
- Explanations: link temperature change to internal energy (random kinetic energy of the molecules), and state that work is done by the gas only when its volume increases.

SECTION A TRAPS

Trap: Wrong sign convention: taking W as the work done on the gas while using $\Delta U = Q - W$. If the question defines W as the work done on the gas, then $\Delta U = Q + W$.

Trap: Assuming that heat supplied always raises the temperature; during an isothermal expansion the heat supplied is converted entirely into work and $\Delta U = 0$.

Trap: Using Q instead of ΔU to find the temperature change when the volume or pressure also changes; ΔU depends only on ΔT, and Q depends on the process.

TRICKS

Trick: ΔU depends only on the initial and final temperatures, so for any process between the same two temperatures ΔU is identical; only Q and W differ.

Trick: A quick sign check: gas heated and expands, so Q is positive and W is positive; gas compressed rapidly with no heat flow, so W is negative and ΔU is positive (temperature rises).

11.2 Thermodynamic Processes

DEFINITIONS & LAWS

- **Isothermal process** is a process at constant temperature: $\Delta U = 0$, so $Q = W$; heat absorbed is converted entirely into work done by the gas (pV = constant, Boyle's law).
- **Adiabatic process** is a process in which no heat enters or leaves the system: $Q = 0$, so $\Delta U = -W$; an adiabatic expansion cools the gas and an adiabatic compression heats it.
- **Constant-pressure (isobaric) process**: $W = p\Delta V$, and $\Delta U = Q - p\Delta V$. **Constant-volume (isochoric) process**: $W = 0$, so $Q = \Delta U$, and all the heat supplied increases the internal energy.
- Work done by the gas equals the area under the p–V graph; in a cycle, the net work done equals the area enclosed by the loop (clockwise: net work done by the gas; anticlockwise: net work done on the gas).
- An adiabatic curve is steeper than an isothermal curve through the same point on a p–V graph because γ is greater than 1.
- **Reversible process** is one in which the system passes through a continuous series of equilibrium states and can be returned to the initial state with no change in the surroundings.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Isothermal process	$W = nRT \ln\frac{V_2}{V_1} = p_1V_1 \ln\frac{V_2}{V_1}$; pV = constant	W = work done by the gas (J); n = number of moles; T = constant temperature (K); V_1, V_2 = initial and final volumes (m^3); p_1 = initial pressure (Pa)
Adiabatic relations	pV^γ = constant; $TV^{\gamma-1}$ = constant; $T^\gamma p^{1-\gamma}$ = constant	$\gamma = \frac{C_p}{C_V}$ ($\frac{5}{3}$ monatomic, $\frac{7}{5}$ diatomic)
Adiabatic work	$W = \frac{p_1V_1 - p_2V_2}{\gamma - 1} = \frac{nR(T_1 - T_2)}{\gamma - 1} = -\Delta U$	W = work done by the gas (J); p_1, V_1, T_1 = initial state; p_2, V_2, T_2 = final state; n = number of moles; R = molar gas constant (J mol^{-1} K^{-1}); $\gamma = \frac{C_p}{C_V}$
Constant-pressure (isobaric) process	$W = p\Delta V = nR\Delta T$; $Q = nC_p\Delta T$	constant pressure; p = pressure (Pa); ΔV = volume change (m^3); ΔT = temperature change (K); C_p = molar heat capacity at constant pressure (J mol^{-1} K^{-1})
Constant-volume (isochoric) process	$W = 0$; $Q = \Delta U = nC_V\Delta T$	constant volume; C_V = molar heat capacity at constant volume (J mol^{-1} K^{-1}); ΔT = temperature change (K)
Mayer's relation	$C_p - C_V = R$	C_p, C_V = molar heat capacities at constant pressure and volume (J mol^{-1} K^{-1}); R = molar gas constant (J mol^{-1} K^{-1})

Quantity / law	Formula	Symbols, units & conditions
First law for each process	$Q = \Delta U + W$	Q = heat supplied to the gas (J); ΔU = change in internal energy (J); W = work done by the gas (J)

HOW TO ANSWER (MPM STYLE)

- For each process, state the defining condition ($\Delta T = 0$, $Q = 0$, $\Delta p = 0$ or $\Delta V = 0$), then write the reduced form of the first law before substituting.
- Calculation: use $pV = nRT$ to find the missing p, V or T, then W from the area or formula, $\Delta U = \frac{f}{2} nR\Delta T$, and $Q = \Delta U + W$; show every step with units.
- p–V sketch: label axes, mark the initial and final states with arrows, draw a horizontal line (constant pressure), a vertical line (constant volume), a hyperbola (isothermal) and a steeper curve (adiabatic).
- For a cycle, tabulate Q, W and ΔU for each stage; ΔU over a complete cycle is zero, so the net heat supplied equals the net work done.

SECTION A TRAPS

Trap: Confusing isothermal and adiabatic: isothermal means $\Delta U = 0$ with Q and W non-zero; adiabatic means $Q = 0$ with ΔU and W non-zero.

Trap: Applying pV^{γ} = constant to an isothermal process, or $W = nRT \ln\frac{V_2}{V_1}$ to an adiabatic process; each formula belongs to one process only.

Trap: Taking $W = p\Delta V$ for a process in which the pressure changes; this holds only for constant-pressure changes, otherwise use the area under the graph.

TRICKS

Trick: Reading a p–V graph: the horizontal line is at constant pressure, the vertical line is at constant volume, and of two curves from the same point the steeper one is adiabatic.

Trick: At the same volume change from the same start, an adiabatic expansion ends at a lower pressure and a lower temperature than an isothermal expansion.

Chapter 12

TRANSFER OF THERMAL ENERGY

12.1 Conduction, Convection and Radiation

DEFINITIONS & LAWS

- **Conduction** is the transfer of heat through a material without bulk movement of the material, by lattice vibrations of atoms and, in metals, by the collisions of free electrons; it needs a medium.
- **Convection** is the transfer of heat by the bulk movement of a fluid (liquid or gas) caused by density differences; **radiation** is the transfer of heat by electromagnetic waves (mainly infrared) and does not need a medium.
- **Thermal conductivity** k is the rate of heat flow through a unit area of a material per unit temperature gradient, i.e. the heat transferred per unit time across unit area when the temperature gradient is 1 K m^{-1}.
- **Stefan–Boltzmann law**: the total power radiated per unit surface area of a black body is directly proportional to the fourth power of its thermodynamic temperature.
- A **black body** is a perfect absorber and a perfect emitter of radiation at all wavelengths (emissivity e = 1).
- **Steady state**: the temperature at each point does not change with time, so the rate of heat flow is the same through every cross-section in series.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Rate of heat conduction	$\frac{Q}{t} = \frac{kA(\theta_1 - \theta_2)}{L} = -kA\frac{d\theta}{dx}$	$\frac{Q}{t}$ = rate of heat flow (W); k = thermal conductivity (W m^{-1} K^{-1}); A = cross-sectional area (m^2); L = length or thickness (m); θ_1, θ_2 = temperatures of the two faces; $\frac{d\theta}{dx}$ = temperature gradient (K m^{-1})
Thermal resistance	$R = \frac{L}{kA}$	R = thermal resistance (K W^{-1}); L = length or thickness (m); k = thermal conductivity (W m^{-1} K^{-1}); A = cross-sectional area (m^2)
Layers in series	$R = R_1 + R_2 + \ldots$	R = total thermal resistance (K W^{-1}); R_1, R_2, … = thermal resistances of the layers
Heat flow through thermal resistance	$\frac{Q}{t} = \frac{\Delta\theta}{R}$	Δθ = temperature difference (K); R = thermal resistance (K W^{-1})

Quantity / law	Formula	Symbols, units & conditions
Power radiated	$P = e\sigma AT^4$	P = power radiated (W); $\sigma = 5.67\times10^{-8}$ W m^{-2} K^{-4}; e = emissivity; A = surface area (m^2); T = temperature (K, in kelvin)
Net power radiated	$P_{net} = e\sigma A(T^4 - T_0^4)$	body at temperature T in surroundings at temperature T_0 (K); P_{net} = net power radiated (W); e = emissivity; A = surface area (m^2); σ = Stefan–Boltzmann constant
Surface area of a sphere	$A = 4\pi r^2$	for a spherical star or planet; r = radius (m); A = surface area (m^2)
Intensity at a distance	$I = \frac{P}{4\pi d^2}$	I = intensity at distance d (W m^{-2}); P = power emitted (W); d = distance from the source (m)

- Energy balance: at equilibrium, power absorbed = power emitted.

HOW TO ANSWER (MPM STYLE)

- Conduction: state $\frac{Q}{t} = \frac{kA\Delta\theta}{L}$, then substitute with L in metres and Δθ in K or °C (a difference only), and give the answer in W.
- Composite bar or wall in series: state that the rate of heat flow is the same through each layer in steady state, and write the interface temperature from $\frac{Q}{t}$ equal for the two sections.
- Radiation: convert T to K before using T^4, include e (equal to 1 for a black body), and use the $T^4 - T_0^4$ form for net power.
- For a qualitative explanation, link each mode to its mechanism: free electrons (metals, good conductors), density difference (convection currents), and electromagnetic waves (no medium; the Sun heats the Earth through vacuum).

SECTION A TRAPS

Trap: Using °C in $P = e\sigma AT^4$; the temperature must be in kelvin, but in Δθ for conduction the temperature difference is the same in °C and K.

Trap: Stating that the heat flow differs between layers of different materials in series at steady state; the rate of flow is the same, and it is the temperature gradient that is larger in the poorer conductor.

Trap: Forgetting the fourth power: doubling the thermodynamic temperature increases the power by a factor of 16, not 2 or 4; and using the radius in place of the area $A = 4\pi r^2$ for a sphere.

TRICKS

Trick: Ratios: for the same bar, $\frac{Q}{t} \propto A$ and $\propto \frac{1}{L}$; doubling the radius of a rod gives a 4 times larger heat flow, and doubling the length halves it.

Trick: Radiation ratio: $P \propto AT^4$, so compare two bodies with $\frac{A_1}{A_2}(\frac{T_1}{T_2})^4$ and cancel σ and e when they are the same.